

The Condensed Textbook

Definitions and model proofs from Epp, *Discrete Mathematics with Applications* (5th ed.), scoped to Lectures 1–6

About this book

Definitions follow the wording of the lecture slides. Proofs follow the lecture format: numbered steps, each with its justification in brackets. Worked problems come from the AY23/24, AY24/25 and AY25/26 midterms, and the answers agree with the official answer sheets.

Box types

DEFINITION

A definition, in the lecture wording.

THEOREM

A result that may be cited in a proof.

PROOF

A model proof in exam format.

PAST PAPER

A past midterm question with its solution.

TRAP

A common error.

METHOD

A procedure for a question type.

PRACTICE

A new exam-style problem; its model solution follows.

WORKED EXAMPLE

A lecture or Epp example; **Drill** boxes are quick self-checks.

Format of the test

- 90 minutes, **open book**, 50 marks.
- **Part A:** 15 MCQs $\times$ 2 marks = 30. One answer each; many are “which of (i)–(iv) are true”.
- **Part B:** 20 marks written: set and relation computations, a short proof or two, occasionally a logic puzzle.

Topics in the last three papers

Topic	AY23	AY24	AY25
Propositional logic, arguments	2	3	3
Quantified statements	3	1	2
Number theory	0	0	1
Sets, power sets, partitions	4	4	3
Relations: composition, properties, closures	3	6	5
Equivalence relations, $A/\sim$	2	1	5
Partial orders, Hasse, chains	4	7	1
Items counted	18	22	20

Each MCQ and each Part B sub-part is counted once, under its main topic; the course-trivia Q1s are excluded. Relations and partial orders appear in Part B every year.

CS1231S conventions (differ from Epp)

Here	Meaning / Epp equivalent
$\mathbb{N}$	$\{0, 1, 2, \dots\}$, including 0
$\overline{A}$	complement (Epp: A^c)
$A \setminus B$	difference (Epp: $A - B$)
$A \subseteq B$	subset; avoid $\subset$
largest / smallest	Epp: greatest / least
$\sim, \rightarrow, \leftrightarrow$	not, implies, iff

Contents. Part I Propositional Logic (Epp Ch. 2) · Part II Quantified Statements (Ch. 3) · Part III Number Theory and Methods of Proof (Ch. 1, 4) · Part IV Set Theory (Ch. 1.2, 6) · Part V Relations (Ch. 1.3, 8, plus the lecture-only material) · Part VI Practice with Model Solutions · Part VII Exam Playbook and Definition Index.

Part I · Propositional Logic

1 · Statements, connectives, truth tables

DEFINITION Statement (2.1.1) · Statement form (2.1.5)

A **statement** (proposition) is a sentence that is true or false, but not both. A **statement form** is an expression built from statement variables and connectives that becomes a statement when statements are substituted for its variables.

p	q	$\sim p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

Order of operations. $\sim$ first; then $\wedge$ and $\vee$, which are **coequal**, so $p \wedge q \vee r$ is ambiguous and must be bracketed; then $\rightarrow, \leftrightarrow$ last. (CS2100 puts $\wedge$ before $\vee$; CS1231S does not.)

Exclusive or: $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$.

$A \rightarrow$ is false in exactly one row: **T** $\rightarrow$ **F**.

DEFINITION Tautology, contradiction (2.1.7–8) · Logical equivalence (2.1.6)

A **tautology** is a statement form that is true for every assignment; a **contradiction** is false for every assignment. $P \equiv Q$ iff P and Q have identical truth values under every assignment; equivalently, iff $P \leftrightarrow Q$ is a tautology.

2 · Logical equivalences and set identities

Theorem 2.1.1 (logic) and Theorem 6.2.2 (sets) side by side, as in Epp's Table 6.4.1. Read $\wedge$ as $\cap$, $\vee$ as $\cup$, $\sim$ as complement, t as U , c as $\emptyset$. Cite laws **by name**.

Law	Logic (for statement variables)	Sets (subsets of U)
Commutative	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$	$A \cap B = B \cap A$ $A \cup B = B \cup A$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$p \wedge t \equiv p$ $p \vee c \equiv p$	$A \cap U = A$ $A \cup \emptyset = A$
Negation / Complement	$p \vee \sim p \equiv t$ $p \wedge \sim p \equiv c$	$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$
Double negation / complement	$\sim(\sim p) \equiv p$	$\overline{\bar{A}} = A$
Idempotent	$p \wedge p \equiv p$ $p \vee p \equiv p$	$A \cap A = A$ $A \cup A = A$
Universal bound	$p \vee t \equiv t$ $p \wedge c \equiv c$	$A \cup U = U$ $A \cap \emptyset = \emptyset$
De Morgan's	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\overline{A \cap B} = \bar{A} \cup \bar{B}$
	$\sim(p \vee q) \equiv \sim p \wedge \sim q$	$\overline{A \cup B} = \bar{A} \cap \bar{B}$
Absorption	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$
Negations of t, c	$\sim t \equiv c$ $\sim c \equiv t$	$\bar{U} = \emptyset$ $\bar{\emptyset} = U$
Set difference	—	$A \setminus B = A \cap \bar{B}$

THEOREM Further equivalences

- **Implication law:** $p \rightarrow q \equiv \sim p \vee q$
- **Contrapositive:** $p \rightarrow q \equiv \sim q \rightarrow \sim p$
- **Negated conditional:** $\sim(p \rightarrow q) \equiv p \wedge \sim q$
- **Biconditional:** $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- **Variant absorption:** $p \wedge (\sim p \vee q) \equiv p \wedge q$
- **Variant absorption:** $p \vee (\sim p \wedge q) \equiv p \vee q$
- **Exportation:** $(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$
- **Cases:** $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$
- $p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$

The left column is proved in Lecture 2. The right column comes from tutorials and assignments; each follows from the table in a line or two, so derive it if a question says “use the laws” (the AY24 answer key cites variant absorption as an assignment result).

PAST PAPER AY24 Q4 · Simplify $((p \rightarrow q) \rightarrow (q \rightarrow r)) \rightarrow (r \rightarrow s) \vee p \wedge (p \rightarrow (p \rightarrow q)) \wedge p$

Call the long first conjunct $X \vee p$. Answer: $p \wedge q$ (option C).

$$\begin{aligned}
 & (X \vee p) \wedge (p \rightarrow (p \rightarrow q)) \wedge p \\
 \equiv & (X \vee p) \wedge (p \wedge (p \rightarrow (p \rightarrow q))) && \text{(associative, commutative)} \\
 \equiv & ((X \vee p) \wedge p) \wedge (p \rightarrow (p \rightarrow q)) && \text{(associative)} \\
 \equiv & p \wedge (p \rightarrow (p \rightarrow q)) && \text{(commutative } \times 2, \text{ absorption)} \\
 \equiv & p \wedge (\sim p \vee (\sim p \vee q)) && \text{(implication law } \times 2) \\
 \equiv & p \wedge (\sim p \vee q) && \text{(associative, idempotent)} \\
 \equiv & p \wedge q && \text{(variant absorption)}
 \end{aligned}$$

METHOD Checking tautologies and equivalences

To test whether $P \rightarrow Q$ is a tautology, try to make it false: set Q false and P true, and follow the consequences. If this forces a contradiction, $P \rightarrow Q$ is a tautology. **AY24 Q3(iii)** $(p \vee q \vee \sim r) \wedge (s \vee \sim(p \vee q)) \rightarrow (s \vee \sim r)$: falsifying needs $s = F, r = T$; the premise becomes $(p \vee q) \wedge \sim(p \vee q)$, a contradiction, so the statement is a tautology. **To show $P \not\equiv Q$** , one row is enough. AY23 Q3(i): $\sim(p \rightarrow q) \equiv p \vee (p \rightarrow \sim q)$? At $p = q = T$ the left side is F and the right side is T, so they are not equivalent.

3 · Conditional statements

DEFINITION Conditional (2.2.1) and its relatives (2.2.2–4)

$p \rightarrow q$ is false only when p is true and q is false; when p is false it is **vacuously true**. From $p \rightarrow q$ form the **contrapositive** $\sim q \rightarrow \sim p$, the **converse** $q \rightarrow p$, and the **inverse** $\sim p \rightarrow \sim q$.

THEOREM

A conditional $\equiv$ its contrapositive. Its converse $\equiv$ its inverse (they are contrapositives of each other). A conditional is **not** equivalent to its converse or its inverse.

DEFINITION Only if (2.2.5) · Biconditional (2.2.6) · Necessary and sufficient (2.2.7)

“ p only if q ” means $\sim q \rightarrow \sim p$, i.e. $p \rightarrow q$. “ p iff q ” is $p \leftrightarrow q$, true exactly when p, q agree.

“ r is **sufficient** for s ” means $r \rightarrow s$. “ r is **necessary** for s ” means $\sim r \rightarrow \sim s$, i.e. $s \rightarrow r$.

English	Symbolic	English	Symbolic
if p then q ; p implies q	$p \rightarrow q$	p only if q	$p \rightarrow q$
q if p ; q whenever p	$p \rightarrow q$	p unless q	$\sim q \rightarrow p \equiv p \vee q$
p is sufficient for q	$p \rightarrow q$	p is necessary for q	$q \rightarrow p$
p if and only if q	$p \leftrightarrow q$	neither p nor q	$\sim p \wedge \sim q$

TRAP Everyday “if” is often an “iff”

“If you behave, you will get ice-cream” is usually understood as a biconditional. In mathematics it is only $b \rightarrow i$; misbehaving and still getting ice-cream does not make it false.

4 · Arguments and rules of inference

DEFINITION Argument, validity (2.3.1) · Sound (2.3.2)

An **argument** is a sequence of statements; all but the last are **premises**, the last (after $\therefore$) is the **conclusion**. It is **valid** iff whenever all premises are true, the conclusion is true; that is, no row of the truth table has all premises T and the conclusion F (check only the **critical rows**, where all premises are T). An argument is **sound** iff it is valid **and** its premises are true.

Rule	Form	Rule	Form
Modus ponens	$p \rightarrow q, p \therefore q$	Generalization	$p \therefore p \vee q$

Modus tollens	$p \rightarrow q, \sim q \therefore \sim p$	Specialization	$p \wedge q \therefore p$
Conjunction	$p, q \therefore p \wedge q$	Elimination	$p \vee q, \sim q \therefore p$
Transitivity	$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$	Contradiction rule	$\sim p \rightarrow c \therefore p$
Division into cases	$p \vee q, p \rightarrow r, q \rightarrow r \therefore r$		

Fallacies (invalid): *converse error* $p \rightarrow q, q \therefore p$; *inverse error* $p \rightarrow q, \sim p \therefore \sim q$.

PAST PAPER AY24 Q5 · Which conclusions follow from $p \rightarrow (q \wedge r)$, $\sim p \rightarrow \sim r$, $\sim p \wedge (r \rightarrow \sim p)$?

1. $\sim p$ (specialization, premise 3) then $\sim r$ (modus ponens with premise 2).
2. So every critical row has $p = r = F$ (q free). Then (i) $p \rightarrow q$ is T (vacuous); (ii) $\sim p \vee \sim r$ is T; (iii) $q \rightarrow (p \rightarrow r)$ is T since $p \rightarrow r$ is T; (iv) $\sim p \wedge \sim q \wedge r$ is F since $r = F$.

Valid: (i), (ii), (iii). Answer C.

PAST PAPER AY25 Q13 · Sound vs valid

Sound $\Rightarrow$ valid by definition, so “ $\forall x(\text{Sound}(x) \rightarrow \text{Valid}(x))$ ” is **true**; the converse fails (valid with a false premise), so “ $\exists x(\sim \text{Sound}(x) \wedge \text{Valid}(x))$ ” is **true**; the other two are false. Answer: **only I and IV**.

PAST PAPER AY23 Q4 · Knights and knaves

Method: let A, D, N mean “is a knight”. A statement S made by X gives $X \leftrightarrow S$. Here: $A \leftrightarrow D$; $D \leftrightarrow \sim N$; “Aiken is not my type” gives $N \leftrightarrow \sim(A \leftrightarrow N)$; “Aiken lies” gives $N \leftrightarrow \sim A$. From $N \leftrightarrow \sim A$, A and N differ, so $\sim(A \leftrightarrow N)$ is T, hence N is T. Then $A = F$, $D = A = F$, and $D \leftrightarrow \sim N$ checks out. **Aiken and Dueet are knaves; Noasu is a knight.** Answer B.

PAST PAPER AY25 Q16 (6 marks) · Four animals, four colours, nine clues

Names: cat (3 letters), sheep (5), rabbit (6), penguin (7).

1. (2) six letters $\leftrightarrow$ black: **rabbit** is black; cat, sheep, penguin are not.
2. (3) an “e” $\rightarrow$ white: **sheep** and **penguin** are white.
3. (4) a repeated letter $\rightarrow$ not grey: sheep (ee), rabbit (bb), penguin (nn) are not grey. (5) brown $\rightarrow$ has an “i”: cat and sheep are not brown.
4. (7) sheep has exactly one colour: **sheep = {white}**. (9) someone is grey, and only the cat can be: **cat = {grey}** (exactly one, by (7)).
5. (6) penguin has exactly two colours: white, and not black or grey, so **penguin = {white, brown}**.
6. (8) some animal is exactly {brown, black}; only the rabbit is black: **rabbit = {brown, black}**. (1) holds.

Model derivations

PROOF Simplify $\sim(\sim p \wedge q) \wedge (p \vee q)$ using the laws (Epp 2.1)

$$\begin{aligned}
 &\sim(\sim p \wedge q) \wedge (p \vee q) \equiv (\sim\sim p \vee \sim q) \wedge (p \vee q) && \text{(De Morgan's law)} \\
 &\equiv (p \vee \sim q) \wedge (p \vee q) && \text{(double negative law)} \\
 &\equiv p \vee (\sim q \wedge q) && \text{(distributive law)} \\
 &\equiv p \vee (q \wedge \sim q) \equiv p \vee c && \text{(commutative law; negation law)} \\
 &\equiv p && \text{(identity law)}
 \end{aligned}$$

■

PROOF Derive variant absorption $p \wedge (\sim p \vee q) \equiv p \wedge q$ from the table

$$p \wedge (\sim p \vee q) \equiv (p \wedge \sim p) \vee (p \wedge q) \quad \text{(distributive)} \equiv c \vee (p \wedge q) \quad \text{(negation)} \equiv (p \wedge q) \vee c \quad \text{(commutative)} \equiv p \wedge q \quad \text{(identity)}.$$

■

PROOF A deduction with named rules: from $p \vee q$, $q \rightarrow r$, $(p \wedge s) \rightarrow t$, $\sim r$, $\sim q \rightarrow (u \wedge s)$, conclude t

1. $\sim q$ (from $q \rightarrow r$ and $\sim r$, by modus tollens)
2. $u \wedge s$ (from $\sim q \rightarrow (u \wedge s)$ and step 1, by modus ponens) so s (specialization)
3. p (from $p \vee q$ and step 1, by elimination)
4. $p \wedge s$ (steps 2, 3, by conjunction) so t (from $(p \wedge s) \rightarrow t$, by modus ponens).

■

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

METHOD Validity by critical rows

For $p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$ (transitivity): the **critical rows** are those where every premise is T: rows 1, 5, 7 and 8 (bold). The conclusion is T in each, so the argument is **valid**. A critical row with a false conclusion would make the argument invalid; that row is a counterexample.

TRAP $\rightarrow$ is not associative

$(p \rightarrow q) \rightarrow r \not\equiv p \rightarrow (q \rightarrow r)$: at $p = q = r = F$ the left is $T \rightarrow F = F$, the right is T . Always bracket chains of $\rightarrow$.

Part II · Quantified Statements

5 · Predicates and quantifiers

DEFINITION Predicate (3.1.1) · Truth set (3.1.2)

A **predicate** is a sentence with finitely many variables that becomes a statement when values are substituted. The **domain** of a variable is the set of values that may be substituted for it. The **truth set** of $P(x)$, $x \in D$, is $\{x \in D : P(x)\}$.

DEFINITION Universal (3.1.3) and existential (3.1.4) statements · Uniqueness

$\forall x \in D, Q(x)$ is true iff $Q(x)$ is true for **every** $x \in D$, and false iff $Q(x)$ is false for at least one x , called a **counterexample**.

$\exists x \in D$ such that $Q(x)$ is true iff $Q(x)$ is true for **at least one** $x \in D$, and false iff $Q(x)$ is false for all $x \in D$.

$\exists! x \in D, Q(x) \equiv \exists x \in D (Q(x) \wedge \forall y \in D (Q(y) \rightarrow y = x))$, read “there is exactly one”.

METHOD Translating English

Use $\rightarrow$ under $\forall$ and $\wedge$ under $\exists$. “All A are B ” is $\forall x(A(x) \rightarrow B(x))$; “some A is B ” is $\exists x(A(x) \wedge B(x))$.

Writing $\forall x(A(x) \wedge B(x))$ claims **everything** is an A ; writing $\exists x(A(x) \rightarrow B(x))$ is true as soon as one non- A exists.

Implicit quantification: “If $x > 2$ then $x^2 > 4$ ” in a real-number context means $\forall x \in \mathbb{R}(x > 2 \rightarrow x^2 > 4)$.

PAST PAPER AY24 Q2 · “All birds cannot fly”

$\forall x(Bird(x) \rightarrow \sim Fly(x))$, option C. Option B, $\forall x(Bird(x) \wedge \sim Fly(x))$, asserts that everything is a bird; D quantifies a second, unrelated y .

6 · Negation, vacuous truth, variants

THEOREM Negation of quantified statements (3.2.1–2)

$\sim(\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x)$ $\sim(\exists x \in D, P(x)) \equiv \forall x \in D, \sim P(x)$

Universal conditional: $\sim\forall x \in D(P(x) \rightarrow Q(x)) \equiv \exists x \in D(P(x) \wedge \sim Q(x))$.

Nested: push $\sim$ inward, flipping each quantifier: $\sim\forall x\exists y P(x, y) \equiv \exists x\forall y \sim P(x, y)$.

DEFINITION Vacuous truth

$\forall x \in D(P(x) \rightarrow Q(x))$ is **vacuously true** when no $x \in D$ satisfies $P(x)$. In particular, every universal statement over an **empty** domain is true, and every existential statement over an empty domain is false.

DEFINITION Variants (3.2.1) · Necessary, sufficient, only if (3.2.2)

For $\forall x \in D(P(x) \rightarrow Q(x))$: **contrapositive** $\forall x(\sim Q(x) \rightarrow \sim P(x))$ (equivalent), **converse** $\forall x(Q(x) \rightarrow P(x))$, **inverse** $\forall x(\sim P(x) \rightarrow \sim Q(x))$.

“ $r(x)$ is sufficient for $s(x)$ ”: $\forall x(r(x) \rightarrow s(x))$. “ $r(x)$ is necessary for $s(x)$ ”: $\forall x(s(x) \rightarrow r(x))$. “ $r(x)$ only if $s(x)$ ”: $\forall x(r(x) \rightarrow s(x))$.

PAST PAPER AY23 Q2 · Given $\forall x \in S(SoC(x) \wedge CS1231(x))$ is true

(i) All computing students took CS1231S: **true**. (ii) Every student is a computing student: **true**, because of the $\wedge$. (iii) $S \neq \emptyset$: **false**, since the statement is vacuously true when $S = \emptyset$. (iv) Some non-computing student took it: **false**, as there is no such student. Only (i) and (ii) hold; no option lists this, so the answer is **E**.

PAST PAPER AY25 Q2 · $\forall x \in D(Red(x) \wedge Flying(x))$ for the objects in LT11

A: if the statement is true, every object is red. True. B: if true, every object flies. True. C: it is true when LT11 is empty. True, vacuously. D: if it is false, there is a counterexample object, so LT11 is non-empty. True. Options A–D are all correct, so the false option is **E**.

7 · Multiple quantifiers

THEOREM Order of quantifiers

Adjacent quantifiers **of the same kind** commute: $\forall x \forall y \equiv \forall y \forall x$ and $\exists x \exists y \equiv \exists y \exists x$. Swapping a $\forall$ past an $\exists$ usually changes the meaning: in $\forall x \exists y P(x, y)$ the witness y **may depend on** x ; in $\exists y \forall x P(x, y)$ one y must work for all x . $\exists y \forall x P \Rightarrow \forall x \exists y P$, never conversely in general.

METHOD Evaluating and proving nested statements

Read the quantifiers from left to right. A $\forall$ variable may take any value; an $\exists$ variable is chosen afterwards and may depend on the values chosen before it.

To **prove** $\forall x \exists y P$: “Let x be arbitrary. Let $y = f(x)$. Then $P(x, f(x))$ because ...”. To **disprove** it, prove the negation $\exists x \forall y \sim P$: give one x and show that every y fails for it.

PAST PAPER AY23 Q5 · Domain $\mathbb{R}$

- (i) $\forall x \exists y(xy = 1)$ is **false**: at $x = 0$, $0 \cdot y = 0 \neq 1$ for every y .
- (ii) $\exists x \forall y(x^2 \geq y^2)$ is **false**: for any x , $y = |x| + 1$ gives $y^2 > x^2$.
- (iii) $\exists x \forall y \exists z(xy^2 = yz)$ is **true**: take $x = 0$; for any y choose $z = 0$. (Or $x = 1$, $z = y$.) Answer **B**.

PAST PAPER AY23 Q16 (6 marks) · Solving for the witnesses

- (a) $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$; smallest positive integer that can lie in C with $\forall x \in A \forall y \in B \forall z \in C(|x - y| \leq |y - z|)$. For each y the worst x is 1, so we need $|y - z| \geq y - 1$ for $y = 4, 5, 6$: at $z = 1$, $|y - 1| = y - 1$. So $z = 1$.
- (b) $\exists y \in \mathbb{Z}^+ \forall x \in \{10, \dots, 50\} \forall z \in \{3, 5, 7, 9\} (x \leq y^2 \wedge y \leq z^2)$: need $y^2 \geq 50$, so $y \geq 8$, and $y \leq 3^2 = 9$. So $y \in \{8, 9\}$.
- (c) $\forall x \in \mathbb{Z} \exists y \in \mathbb{R} \setminus \{0\} (xy < 1)$: take $y = -1$ if $x \geq 0$, $y = 1$ if $x < 0$; then $xy = -x \leq 0$ or $xy = x < 0$, both < 1 . (The key lists several others, e.g. $y = 1$ at $x = 0$ and $y = -x$ otherwise.)

TRAP Hidden domain changes

$\forall x \in \mathbb{Z}^+ \exists y \in \mathbb{Z} (x + y = 0)$ is true but becomes false over $y \in \mathbb{Z}^+$. Check every domain again after negating or reordering quantifiers.

8 · Arguments with quantified statements

Rule	Form	Rule	Form
Universal instantiation	$\forall x \in D P(x); a \in D \therefore P(a)$	Universal generalization	$P(a)$ for an arbitrary $a \in D \therefore \forall x \in D P(x)$

Universal modus ponens	$\forall x(P(x) \rightarrow Q(x)); P(a) \therefore Q(a)$	Universal modus tollens	$\forall x(P(x) \rightarrow Q(x)); \sim Q(a) \therefore \sim P(a)$
Existential instantiation	$\exists x P(x) \therefore P(c)$ for some new c	Existential generalization	$P(a); a \in D \therefore \exists x \in D P(x)$

Invalid: quantified **converse error** $\forall x(P(x) \rightarrow Q(x)); Q(a) \therefore P(a)$ and **inverse error** $\forall x(P(x) \rightarrow Q(x)); \sim P(a) \therefore \sim Q(a)$.

Validity with diagrams: draw “all P are Q ” as the P -disc inside the Q -disc; the argument is valid iff **every** consistent drawing forces the conclusion.

PAST PAPER AY25 Q6 (number theory in quantifier form) · n a perfect square

(i) “ n is prime” is false ($4 = 2^2$). (ii) “ n has an odd number of positive factors” is true for $n \geq 1$ (divisors pair up as $d \leftrightarrow n/d$, except $d = \sqrt{n}$), but **$n = 0$ is a perfect square with infinitely many factors**, so the key accepted both C and D. (iii) “ $p \mid n \Rightarrow p^2 \mid n$ ” is true: if $n = m^2$ and $p \mid m^2$ then $p \mid m$ (unique factorization), so $p^2 \mid m^2$.

Model solutions with nested quantifiers

PROOF Negating the definition of a limit (Lecture 1)

$\lim_{n \rightarrow \infty} a_n = L$ means $\forall \varepsilon \in \mathbb{R}^+ \exists N \in \mathbb{Z} \forall n \in \mathbb{Z} (n > N \rightarrow L - \varepsilon < a_n < L + \varepsilon)$.

Its negation, one quantifier at a time: $\exists \varepsilon \in \mathbb{R}^+ \forall N \in \mathbb{Z} \exists n \in \mathbb{Z} (n > N \wedge (a_n \leq L - \varepsilon \vee a_n \geq L + \varepsilon))$. ■

PROOF $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (x + y = 0)$ is true; $\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (x + y = 0)$ is false

- Let x be an arbitrary integer and let $y = -x$. Then $y \in \mathbb{Z}$ (*closure*) and $x + y = 0$.
- For the second, prove the negation $\forall y \in \mathbb{Z} \exists x \in \mathbb{Z} (x + y \neq 0)$: given any y , take $x = 1 - y$; then $x + y = 1 \neq 0$. ■

English	Symbolic ($L(x, y)$: x loves y)
Everybody loves somebody.	$\forall x \exists y L(x, y)$
Somebody is loved by everybody.	$\exists y \forall x L(x, y)$
Nobody loves everybody.	$\sim \exists x \forall y L(x, y) \equiv \forall x \exists y \sim L(x, y)$
There is exactly one person everybody loves.	$\exists! y \forall x L(x, y)$

PROOF Formal negation of a nested quantified conditional (Lecture 3, Tarski’s world)

“There is a square x such that for all triangles y , x is to the right of y ”: $\exists x (Square(x) \wedge \forall y (Triangle(y) \rightarrow RightOf(x, y)))$. Negate step by step:

$$\begin{aligned}
 & \sim \exists x (Square(x) \wedge \forall y (Triangle(y) \rightarrow RightOf(x, y))) \\
 & \equiv \forall x \sim (Square(x) \wedge \forall y (Triangle(y) \rightarrow RightOf(x, y))) && \text{(negation of } \exists \text{)} \\
 & \equiv \forall x (\sim Square(x) \vee \sim \forall y (Triangle(y) \rightarrow RightOf(x, y))) && \text{(De Morgan)} \\
 & \equiv \forall x (\sim Square(x) \vee \exists y \sim (Triangle(y) \rightarrow RightOf(x, y))) && \text{(negation of } \forall \text{)} \\
 & \equiv \forall x (\sim Square(x) \vee \exists y (Triangle(y) \wedge \sim RightOf(x, y))) && \text{(negated conditional)}
 \end{aligned}$$

“Every object is either not a square, or has some triangle it is not to the right of.” ■

PROOF Universal modus ponens inside an ordinary proof (Lecture 3.4.3)

In “the sum of two even integers is even”, each step is an instance of universal modus ponens:

- “If an integer is even, it equals twice some integer; m is a particular even integer; $\therefore m = 2r$ for some integer r .”
- “For all u, v : if u, v are integers then $u + v$ is an integer; r, s are integers; $\therefore r + s$ is an integer.”
- “If a number equals twice some integer, it is even; $2(r + s)$ is such a number; $\therefore m + n$ is even.” ■

Writing “by definition of even” or “by closure” is shorthand for exactly these instantiations.

DRILL Translate, then negate (answers after the arrow)

- “Every student has a friend”: $\forall s \exists f F(s, f) \rightarrow$ negation $\exists s \forall f \sim F(s, f)$.
- “No integer is both even and odd”: $\forall n \sim (E(n) \wedge O(n)) \rightarrow$ negation $\exists n (E(n) \wedge O(n))$.
- “All primes greater than 2 are odd”: $\forall p (P(p) \wedge p > 2 \rightarrow O(p)) \rightarrow \exists p (P(p) \wedge p > 2 \wedge \sim O(p))$.
- “ x is a **minimal** element”: $\forall y (y \preceq x \rightarrow y = x) \rightarrow \exists y (y \preceq x \wedge y \neq x)$.

Part III · Number Theory and Methods of Proof

9 · Writing proofs

Term	Meaning (Lecture 1)
Definition	precise meaning of a term: all and only the required properties
Axiom	assumed true without proof (Peano, Euclid’s postulates)
Theorem	a major result, proved rigorously
Lemma	a small result used to prove a theorem
Corollary	a simple deduction from a theorem
Conjecture	believed true, unproved (Goldbach)

METHOD Proof format

- **Number the steps;** nest sub-arguments (1.1, 1.2, ...).
- **Justify every step** in brackets: “by definition of odd”, “by closure of integers under +”, “by basic algebra”, “by Theorem 4.4.3”.
- Open a universal proof with “Let x be a **particular but arbitrarily chosen** element of D ”.
- Give each existential witness a **fresh** name.
- End with the claim, restated, then ■.

DEFINITION Basic properties of integers (Lecture 1, Appendix A)

For all $x, y, z \in \mathbb{Z}$: **closure** ($x + y, xy \in \mathbb{Z}$), **commutativity**, **associativity**, **distributivity**, **trichotomy** (exactly one of $x < y$, $x = y$, $x > y$). Appendix A’s field and order axioms for $\mathbb{R}$ may also be cited.

Assumption 1: every integer is even or odd, but not both. **Assumption 2:** every rational number can be reduced to lowest terms.

TRAP Common mistakes in proofs (Epp 4.2)

Arguing from examples · reusing a letter ($m = 2r$ and $n = 2r$ makes $m = n$) · assuming what is to be proved · “therefore” without a reason · writing “ $a \mid b = 3$ ” ($a \mid b$ is a statement, not a number).

10 · Definitions and results in number theory

DEFINITION Even, odd · Prime, composite

$n \in \mathbb{Z}$ is **even** $\Leftrightarrow \exists k \in \mathbb{Z} (n = 2k)$; **odd** $\Leftrightarrow \exists k \in \mathbb{Z} (n = 2k + 1)$.
 n is **prime** iff $n > 1$ and $\forall r, s \in \mathbb{Z}^+ (n = rs \rightarrow (r = 1 \wedge s = n) \vee (r = n \wedge s = 1))$.
 n is **composite** iff $n > 1$ and $\exists r, s \in \mathbb{Z}^+ (n = rs \wedge 1 < r < n \wedge 1 < s < n)$. 1 is neither.

DEFINITION Rational, irrational · Lowest terms

$r \in \mathbb{R}$ is **rational** $\Leftrightarrow \exists a, b \in \mathbb{Z} (r = a/b \wedge b \neq 0)$; otherwise **irrational**. A fraction a/b is in **lowest terms** if the largest integer dividing both a and b is 1.

DEFINITION Divisibility

For $n, d \in \mathbb{Z}$: $d \mid n \Leftrightarrow \exists k \in \mathbb{Z} (n = dk)$. Read: d divides n ; n is a multiple of d ; d is a factor/divisor of n .

Consequences: every d divides 0; 0 divides only 0; $1 \mid n$ and $n \mid n$ for all n . No division is performed: $3 \mid 12$ is **true** and $3 \mid 10$ is **false**.

THEOREM Quotient–Remainder Theorem (4.5.1) · div and mod

For $n \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$ there exist **unique** $q, r \in \mathbb{Z}$ with $n = dq + r$ and $0 \leq r < d$. We write $n \operatorname{div} d = q$ and $n \operatorname{mod} d = r$. Hence every integer is of exactly one form $dq, dq + 1, \dots, dq + (d - 1)$. This is the basis for **division into cases**.

THEOREM Results that may be cited

- **4.3.1** Every integer is rational.
- **4.3.2** The sum of two rationals is rational; **Cor.** **4.3.3** the double of a rational is rational.
- **4.4.1** $a, b \in \mathbb{Z}^+, a \mid b \Rightarrow a \leq b$.
- **4.4.2** The only divisors of 1 are 1 and -1 .
- **4.4.3** $a \mid b \wedge b \mid c \Rightarrow a \mid c$.
- **4.4.4** Every integer $n > 1$ is divisible by a prime.
- **4.4.5** Unique factorization: $n > 1$ is $p_1^{e_1} \dots p_k^{e_k}$, unique up to order.
- **4.5.2** Consecutive integers have opposite parity.
- **4.5.4** $-|r| \leq r \leq |r|$; **4.5.6** $|x + y| \leq |x| + |y|$.
- **4.7.1** There is no greatest integer.
- **4.7.4** n^2 even $\Rightarrow n$ even.
- **4.8.1** $\sqrt{2}$ is irrational; **4.8.4** there are infinitely many primes.

Numbers follow Epp's 5th edition. The slides quote 4th-edition numbers first: the slides' 4.k.x is 4.(k+1).x in the 5th edition for sections 4.2–4.7; for example, the slides' "Theorem 4.3.3" is 4.4.3.

11 · Choosing the method

To show	Method	Skeleton
$\exists x \in D P(x)$	Construction	Name a witness $x = \dots$; verify $x \in D$ and $P(x)$. There is no need to explain how it was found.
$\sim \forall x \in D (P(x) \rightarrow Q(x))$	Counterexample	One x with $P(x)$ true and $Q(x)$ false. One suffices.
$\forall x \in D P(x)$, D finite	Exhaustion	Check every case explicitly.
$\forall x \in D (P(x) \rightarrow Q(x))$	Direct (generic particular)	Let $x \in D$ be arbitrary with $P(x)$. Unpack definitions; rebuild $Q(x)$.
same	Division into cases	Split by parity / by $x \operatorname{mod} d$ (QR theorem) / by sign; prove each case.
same	Contraposition	Prove $\forall x (\sim Q(x) \rightarrow \sim P(x))$ directly.
any statement S	Contradiction	Suppose $\sim S$. Derive a contradiction. Conclude S .
$P \Leftrightarrow Q$	Two directions	Prove $P \Rightarrow Q$ and $Q \Rightarrow P$ separately.
$\exists! x P(x)$	Existence + uniqueness	Give x with $P(x)$; then if $P(y)$, show $y = x$.

METHOD Contradiction or contraposition?

Use **contraposition** when the negated conclusion is easier to work with; " n is odd" is easier to use than " n^2 is not odd". Use **contradiction** for statements such as "there is no ...", "there are infinitely many ..." and "... is irrational", whose negations assert that some object exists.

12 · Model proofs

PROOF Sum of two even integers is even (Lecture 4, Ex. 4)

1. Let m and n be particular but arbitrarily chosen even integers.
 - 1.1. Then $m = 2r$ and $n = 2s$ for some integers r and s (by definition of even).
 - 1.2. $m + n = 2r + 2s = 2(r + s)$ (by basic algebra).
 - 1.3. $r + s$ is an integer (by closure of integers under +), so $m + n$ is even (by definition of even).
2. Therefore the sum of any two even integers is even. ■

PROOF The product of two consecutive odd integers is odd (Lecture 1, Ex. 1)

1. Let a and b be two consecutive odd integers.
 - 1.1. WLOG $a < b$, hence $b = a + 2$.
 - 1.2. $a = 2k + 1$ for some integer k (by definition of odd), so $b = 2k + 3$.
 - 1.3. $ab = (2k + 1)(2k + 3) = 4k^2 + 8k + 3 = 2(2k^2 + 4k + 1) + 1$ (by basic algebra).
 - 1.4. Let $m = 2k^2 + 4k + 1$, an integer (by closure under $\times$ and $+$). Then $ab = 2m + 1$ is odd (by definition of odd).
2. Therefore the product of two consecutive odd integers is odd. ■

PROOF Theorem 4.3.2 · The sum of any two rational numbers is rational

1. Let r and s be particular but arbitrarily chosen rational numbers.
 - 1.1. Then $r = a/b$ and $s = c/d$ for some integers a, b, c, d with $b \neq 0, d \neq 0$ (by definition of rational).
 - 1.2. $r + s = (ad + bc)/(bd)$ (by basic algebra).
 - 1.3. $ad + bc$ and bd are integers (by closure under $+$ and $\times$), and $bd \neq 0$ (zero product property), so $r + s$ is rational (by definition of rational).
2. Therefore the sum of any two rational numbers is rational. ■

PROOF Theorem 4.4.1 · For all $a, b \in \mathbb{Z}^+$, if $a \mid b$ then $a \leq b$

1. Let a, b be positive integers with $a \mid b$.
 - 1.1. Then $b = ak$ for some integer k (by definition of divisibility).
 - 1.2. Since a and b are positive, k is positive, i.e. $k \geq 1$.
 - 1.3. Therefore $a \leq ak = b$. ■

PROOF Theorem 4.4.2 · The only divisors of 1 are 1 and -1 (division into cases)

1. Let m be any integer with $m \mid 1$. Then $1 = mk$ for some integer k (by definition of divisibility).
 - 1.1. Since $mk > 0$, m and k are both positive or both negative.
 - 1.2. **Case 1:** both positive. Then $m \leq 1$ (by Theorem 4.4.1), so $m = 1$.
 - 1.3. **Case 2:** both negative. Then $1 = (-m)(-k)$, so $-m$ is a positive divisor of 1; as in Case 1, $-m = 1$, i.e. $m = -1$.
2. Therefore the only divisors of 1 are 1 and -1 . ■

PROOF Theorem 4.4.3 · Transitivity of divisibility

1. Suppose a, b, c are integers with $a \mid b$ and $b \mid c$.
 - 1.1. Then $b = ar$ and $c = bs$ for some integers r, s (by definition of divisibility).
 - 1.2. $c = bs = (ar)s = a(rs)$ (by substitution and associativity).
 - 1.3. rs is an integer (by closure under $\times$), so $a \mid c$ (by definition of divisibility). ■

PROOF Every integer n has $n^2 \bmod 4 \in \{0, 1\}$ (division into cases, via parity)

1. Let n be any integer. n is even or odd (Assumption 1).
 - 1.1. **Case 1:** $n = 2k$ for some integer k . Then $n^2 = 4k^2 = 4(k^2) + 0$, and $k^2 \in \mathbb{Z}$ (closure).
 - 1.2. **Case 2:** $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$, and $k^2 + k \in \mathbb{Z}$ (closure).
2. In either case $n^2 = 4q + r$ with $r \in \{0, 1\}$, so $n^2 \bmod 4 \in \{0, 1\}$ (by the quotient-remainder theorem, r is unique). ■

PROOF Theorem 4.7.1 · There is no greatest integer (contradiction)

1. Suppose not; that is, there is a greatest integer g , so $g \geq n$ for every integer n .
 - 1.1. Let $G = g + 1$. Then G is an integer (by closure under $+$) and $G > g$.
 - 1.2. This contradicts that g is the greatest integer.
2. Hence the supposition is false, and there is no greatest integer. ■

PROOF Proposition 4.7.4 · For all integers n , if n^2 is even then n is even (contraposition)

1. Contrapositive: for all integers n , if n is odd then n^2 is odd.
2. Let n be an arbitrarily chosen odd integer.
 - 2.1. Then $n = 2k + 1$ for some integer k (by definition of odd).
 - 2.2. $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ (by basic algebra).
 - 2.3. Let $m = 2k^2 + 2k$, an integer (by closure). Then $n^2 = 2m + 1$ is odd (by definition of odd).
3. Therefore, for all integers n , if n^2 is even then n is even. ■

PROOF Theorem 4.8.1 · $\sqrt{2}$ is irrational (contradiction)

1. Suppose not; that is, $\sqrt{2}$ is rational.
 - 1.1. Then $\sqrt{2} = a/b$ for some integers a, b with $b \neq 0$ (by definition of rational).
 - 1.2. Reduce a/b to lowest terms m/n (Assumption 2) then $m^2 = 2n^2$ (by basic algebra).
 - 1.3. So m^2 is even (by definition of even, as $n^2 \in \mathbb{Z}$), hence m is even (by Proposition 4.7.4).
 - 1.4. Let $m = 2k$. Then $4k^2 = 2n^2$, so $n^2 = 2k^2$ is even, hence n is even (by Proposition 4.7.4).
 - 1.5. So 2 divides both m and n , contradicting that m/n is in lowest terms.
2. Therefore $\sqrt{2}$ is irrational. ■

PROOF Theorem 4.7.3 · rational + irrational is irrational (contradiction)

1. Suppose not: some rational r and irrational s have $r + s$ rational.
 - 1.1. Then $r = a/b$ and $r + s = c/d$ with $a, b, c, d \in \mathbb{Z}$, $b, d \neq 0$ (by definition of rational).
 - 1.2. So $s = c/d - a/b = (bc - ad)/(bd)$, with $bc - ad, bd \in \mathbb{Z}$ and $bd \neq 0$ (closure; zero product property).
 - 1.3. Hence s is rational (by definition of rational), contradicting that s is irrational. ■

PROOF Theorem 4.8.4 · The set of primes is infinite (contradiction)

1. Suppose not: the primes are exactly $p_1, p_2, \dots, p_n$. Let $N = p_1 p_2 \cdots p_n + 1$.
 - 1.1. $N > 1$, so some prime p divides N (by Theorem 4.4.4) $p = p_i$ for some i (the list is complete).
 - 1.2. p_i divides $p_1 \cdots p_n = N - 1$ as well. But a prime dividing a cannot divide $a + 1$ (Proposition 4.8.3). Contradiction.
2. Hence there are infinitely many primes. ■

PROOF There exist irrational p, q with p^q rational (Lecture 1, Ex. 7; non-constructive proof by cases)

1. $\sqrt{2}$ is irrational (Theorem 4.8.1). Consider $\sqrt{2}^{\sqrt{2}}$: it is rational or irrational.
 - 1.1. **Case 1:** rational. Take $p = q = \sqrt{2}$.
 - 1.2. **Case 2:** irrational. Take $p = \sqrt{2}^{\sqrt{2}}$, $q = \sqrt{2}$: then $p^q = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$, rational.
2. In either case the required p and q exist. ■

More model proofs

PROOF Theorem 4.5.2 · Any two consecutive integers have opposite parity

1. Let m and $m + 1$ be two particular but arbitrarily chosen consecutive integers. m is even or odd (Assumption 1).
 - 1.1. **Case 1:** $m = 2k$ for some $k \in \mathbb{Z}$. Then $m + 1 = 2k + 1$ is odd (by definition of odd).
 - 1.2. **Case 2:** $m = 2k + 1$. Then $m + 1 = 2(k + 1)$ with $k + 1 \in \mathbb{Z}$ (closure), so $m + 1$ is even.
2. In both cases m and $m + 1$ have opposite parity. ■

PROOF Theorem 4.5.3 · The square of any odd integer has the form $8m + 1$ (division into cases via QR)

1. Let n be a particular but arbitrarily chosen odd integer. By the QR theorem with $d = 4$, $n = 4q, 4q + 1, 4q + 2$ or $4q + 3$ for some $q \in \mathbb{Z}$; as n is odd, $n = 4q + 1$ or $n = 4q + 3$.
 - 1.1. **Case 1:** $n^2 = 16q^2 + 8q + 1 = 8(2q^2 + q) + 1$.
 - 1.2. **Case 2:** $n^2 = 16q^2 + 24q + 9 = 8(2q^2 + 3q + 1) + 1$.
2. In both cases $n^2 = 8m + 1$ for an integer m (closure). ■

PROOF Theorem 4.5.6 · Triangle inequality $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$

1. By Lemma 4.5.4, $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$; adding, $-(|x| + |y|) \leq x + y \leq |x| + |y|$.
 - 1.1. **Case 1:** $x + y \geq 0$. Then $|x + y| = x + y \leq |x| + |y|$ (definition of absolute value).
 - 1.2. **Case 2:** $x + y < 0$. Then $|x + y| = -(x + y) \leq |x| + |y|$ (from the left inequality).
2. So $|x + y| \leq |x| + |y|$ in all cases. ■

PROOF Proposition 4.8.3 · For any integer a and prime p : if $p \mid a$ then $p \nmid (a + 1)$ (contradiction)

1. Suppose not: $p \mid a$ and $p \mid (a + 1)$ for some integer a and prime p .
 - 1.1. Then $a = pr$ and $a + 1 = ps$ for some $r, s \in \mathbb{Z}$ (definition of divisibility), so $1 = ps - pr = p(s - r)$.

1.2. So $p \mid 1$ (as $s - r \in \mathbb{Z}$), hence $p = 1$ or $p = -1$ (Theorem 4.4.2). This is a contradiction, since a prime is greater than 1. ■

PROOF If $a \mid b$ and $a \mid c$ then $a \mid (mb + nc)$ for all integers m, n

1. Suppose $a \mid b$ and $a \mid c$: $b = ar$ and $c = as$ for some $r, s \in \mathbb{Z}$ (definition of divisibility).
2. Then $mb + nc = a(mr + ns)$ (basic algebra), and $mr + ns \in \mathbb{Z}$ (closure), so $a \mid (mb + nc)$. ■

PROOF Lecture 1 · If x and y are “colorful” (n colorful $\leftrightarrow \exists k \in \mathbb{Z} (n = 3k)$), so is $x + 2y$
Work backwards first: the goal “ $x + 2y$ is colorful” means we need an integer c with $x + 2y = 3c$.

1. Let x, y be colorful integers. Then $x = 3a$ and $y = 3b$ for some $a, b \in \mathbb{Z}$ (by definition of colorful).
2. $x + 2y = 3a + 6b = 3(a + 2b)$ (basic algebra). Let $c = a + 2b$, an integer (closure).
3. So $x + 2y = 3c$, and $x + 2y$ is colorful (by definition of colorful). ■

WORKED EXAMPLE Disproof by counterexample

- “ $\forall a, b, c \in \mathbb{Z} (a \mid bc \rightarrow a \mid b \vee a \mid c)$ ” is false: $4 \mid 2 \cdot 2$ but $4 \nmid 2$.
- “ $\forall a, b \in \mathbb{Z} (a \mid b \wedge b \mid a \rightarrow a = b)$ ” is false: take $a = 1, b = -1$. (True on $\mathbb{Z}^+$: Theorem 4.4.1.)
- “The product of two irrational numbers is irrational” is false: $\sqrt{2} \cdot \sqrt{8} = 4$ (Lecture 1).
- “ $\forall a, b \in \mathbb{R} (a^2 = b^2 \rightarrow a = b)$ ” is false: take $a = 1, b = -1$ (Lecture 4).
- “There is no integer that is both even and odd” (4.7.2) is **true**: if $n = 2a = 2b + 1$ then $1 = 2(a - b)$, so $1/2 \in \mathbb{Z}$, a contradiction.

PROOF Theorem 4.3.1 · Every integer is a rational number

1. Let a be a particular but arbitrarily chosen integer. Then $a = a/1$, a quotient of integers with non-zero denominator.
2. Hence a is rational (by definition of rational). ■

PROOF Proof by exhaustion (Lecture 4, Ex. 3) · every even n with $4 \leq n \leq 26$ is a sum of two primes

$4 = 2 + 2, 6 = 3 + 3, 8 = 3 + 5, 10 = 5 + 5, 12 = 5 + 7, 14 = 3 + 11, 16 = 5 + 11, 18 = 7 + 11, 20 = 7 + 13, 22 = 5 + 17, 24 = 5 + 19, 26 = 7 + 19$. Every case is checked. ■

PROOF For every integer n , $3 \mid (n^3 - n)$ (division into cases, QR with $d = 3$)

1. Let $n \in \mathbb{Z}$. By the QR theorem, $n = 3q, 3q + 1$ or $3q + 2$ for some $q \in \mathbb{Z}$. Note $n^3 - n = (n - 1)n(n + 1)$ (basic algebra).
 - 1.1. **Case $n = 3q$:** the factor $n = 3q$, so $n^3 - n = 3 \cdot q(n - 1)(n + 1)$.
 - 1.2. **Case $n = 3q + 1$:** the factor $n - 1 = 3q$, so $n^3 - n = 3 \cdot qn(n + 1)$.
 - 1.3. **Case $n = 3q + 2$:** the factor $n + 1 = 3(q + 1)$, so $n^3 - n = 3 \cdot (q + 1)(n - 1)n$.
2. In each case the cofactor is an integer (closure), so $3 \mid (n^3 - n)$ (definition of divisibility). ■

PROOF Every prime $p > 3$ has the form $6k + 1$ or $6k + 5$ (Epp 4.5, exercise 39)

1. Let $p > 3$ be prime. By the QR theorem, $p = 6k + r$ with $r \in \{0, 1, 2, 3, 4, 5\}$.
 - 1.1. If $r \in \{0, 2, 4\}$, then $p = 2(3k + r/2)$ is even and > 2 , so not prime.
 - 1.2. If $r = 3$, then $p = 3(2k + 1)$ with $p > 3$, so not prime.
2. Hence $r = 1$ or $r = 5$. ■

PROOF Theorem 4.4.4 · Every integer $n > 1$ is divisible by a prime (via well-ordering)

1. Let $n > 1$ be an integer, and let $D = \{d \in \mathbb{Z} : d > 1 \wedge d \mid n\}$. $D \neq \emptyset$, since $n \in D$.
2. D has a smallest element p ($(\mathbb{N}, \leq)$ is well-ordered).
3. p is prime: suppose instead $p = rs$ with $1 < r < p$. Then $r \mid p$ and $p \mid n$, so $r \mid n$ (Theorem 4.4.3), so $r \in D$ with $r < p$, contradicting the minimality of p .
4. Hence the prime p divides n . ■

PROOF Lemma 4.5.4 · For every real r : $-|r| \leq r \leq |r|$ (division into cases)

1. **Case $r \geq 0$:** $|r| = r$ (definition of absolute value), and $-|r| = -r \leq 0 \leq r$. So $-|r| \leq r = |r|$.

2. **Case $r < 0$:** $|r| = -r > 0$, so $-|r| = r$ and $r < 0 < |r|$. So $-|r| = r \leq |r|$. ■

PAST PAPER Using unique factorization (Theorem 4.4.5) · **AY25 Q6(iii)** in full

Claim: if $n = m^2$ is a perfect square and a prime p divides n , then $p^2 \mid n$.

If $m = 0$, then $n = 0$ and $p^2 \mid 0$. Otherwise write $|m| = p_1^{e_1} \cdots p_k^{e_k}$ (Theorem 4.4.5) then $n = p_1^{2e_1} \cdots p_k^{2e_k}$. A prime dividing n is some p_i (uniqueness of the factorization), and its exponent $2e_i \geq 2$, so $p_i^2 \mid n$. ■

In a perfect square every prime appears to an even power.

Part IV · Set Theory

13 · Describing sets

DEFINITION Set · Membership · Cardinality

A **set** is an unordered collection of objects, its **elements**; order and repetition do not matter:

$\{9, 8, 7\} = \{7, 8, 7, 9, 9\}$. $x \in S$ means x is an element of S . The **cardinality** $|S|$ is the number of elements of S .

DEFINITION Set-builder and replacement notation

Set-builder: $\{x \in U : P(x)\}$ is the set of all $x \in U$ for which $P(x)$ is true. z is a member iff $z \in U$ and $P(z)$.

Replacement: $\{t(x) : x \in A\}$ is the set of all objects $t(x)$ as x ranges over A . z is a member iff $t(x) = z$ for **some** $x \in A$.

E.g. $\{x + 1 : x \in \mathbb{Z}_{\geq 0}\} = \mathbb{Z}^+$; $\{x \in \mathbb{Z} : -2 < x < 5\} = \{-1, 0, 1, 2, 3, 4\}$.

Symbol	Set
$\mathbb{N}$	natural numbers $\{0, 1, 2, \dots\}$, including 0
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$	integers, rationals, reals
$\mathbb{Z}^+, \mathbb{Z}^-, \mathbb{Z}_{\geq 0}$	positive, negative, non-negative integers (0 is neither positive nor negative)
$(a, b), [a, b]$	open / closed real intervals; also $(a, b], [a, \infty) \dots$

PAST PAPER **AY25 Q17(b)** · Name these subsets of $\mathbb{N}$

$A = \{0, 1, 3, 7, 15, 31, \dots\}$ in replacement notation:

$A = \{2^n - 1 : n \in \mathbb{N}\}$.

$B = \{3, 9, 15, 21, 27, \dots\}$ in set-builder notation:

$B = \{n \in \mathbb{N} : 3 \mid n \wedge n \text{ is odd}\}$.

(Many correct answers; check the first few terms.)

14 · Subsets, equality and the empty set

DEFINITION Subset · Proper subset · Set equality

$A \subseteq B \Leftrightarrow \forall x(x \in A \rightarrow x \in B)$. $A \not\subseteq B \Leftrightarrow \exists x(x \in A \wedge x \notin B)$.

$A \subsetneq B$ (proper) iff $A \subseteq B$ and $A \neq B$. $A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A \Leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$.

THEOREM 6.2.4 · The empty set

$\emptyset \subseteq A$ for every set A (and $\emptyset$ is unique). **Proof:** $\forall x(x \in \emptyset \rightarrow x \in A)$ is vacuously true, as $x \in \emptyset$ is always false.

A set with exactly one element is a **singleton**.

Claim		Why
$\emptyset \in \{1, 2, 3\}$	F	$\emptyset$ is not listed
$\emptyset \subseteq \{1, 2, 3\}$	T	Thm 6.2.4
$\{2\} \in \{1, 2, 3\}$	F	elements are numbers
$\{2\} \subseteq \{1, 2, 3\}$	T	2 is an element

The $\emptyset$ ladder	$ \cdot $
$\emptyset = \{\}$	0
$\{\emptyset\} = \mathcal{P}(\emptyset)$	1
$\{\emptyset, \{\emptyset\}\} = \mathcal{P}(\mathcal{P}(\emptyset))$	2
$\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$	4

$\{2\} \in \{\{1\}, \{2\}, \{3\}\}$	T	$\{2\}$ is listed
$\{2\} \subseteq \{\{1\}, \{2\}, \{3\}\}$	F	2 is not listed
$\{9\} = \{\{9\}\}$	F	different elements

$\mathcal{P}^4(\emptyset)$	16
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$\emptyset \neq \{\emptyset\}$: the first is empty, the second has one element. $\emptyset \in \{\emptyset\}$ and $\emptyset \subseteq \{\emptyset\}$ are both true.

TRAP $\subseteq$ is transitive but $\in$ is not · **AY23 Q8**

Which hold for all sets? (i) $A \in B \wedge B \subseteq C \Rightarrow A \in C$ is **true**, since A is an element of B and hence of C .

(ii) $A \in B \wedge B \subseteq C \Rightarrow A \subseteq C$ is false: take $A = \{1\}$, $B = C = \{\{1\}\}$, but $1 \notin C$.

(iii), (iv) $A \subseteq B \wedge B \in C \Rightarrow A \in C$ or $A \subseteq C$ are false: take $A = \{x\}$, $B = \{x, y\}$, $C = \{\{x, y\}\}$. Only (i): **A**.

15 · Ordered pairs and Cartesian products

DEFINITION Ordered pair, n -tuple · Cartesian product

$(a, b) = (c, d) \Leftrightarrow a = c \wedge b = d$; likewise for n -tuples componentwise. $(1, 2) \neq (2, 1)$ though $\{1, 2\} = \{2, 1\}$.

$A \times B = \{(a, b) : a \in A \wedge b \in B\}$; $A_1 \times \dots \times A_n = \{(a_1, \dots, a_n) : a_i \in A_i\}$; $A^n = A \times \dots \times A$.

Facts. $|A \times B| = |A| \cdot |B|$. $A \times \emptyset = \emptyset \times A = \emptyset$. $A \times B \neq B \times A$ in general. $(A \times B) \times C \neq A \times B \times C$: elements $((a, b), c)$ versus (a, b, c) . $A \times (B \cup C) = (A \times B) \cup (A \times C)$, and likewise for $\cap$ and $\setminus$.

PAST PAPER **AY24 Q6** · Which hold for all sets A, B, C ?

(i) $\mathcal{P}(A) \subseteq \mathcal{P}(B) \rightarrow A \subseteq B$ is **true** (proof in §17). (ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ is **true** (proof in §17).

(iii) $A \times B \subseteq B \times C \rightarrow A \subseteq C$ is **false**: take $B = \emptyset$; then $A \times \emptyset = \emptyset$ is a subset of every set, whatever A is.

Answer **A**.

16 · Operations on sets

DEFINITION Union, intersection, difference, complement

For $A, B \subseteq U$: $A \cup B = \{x \in U : x \in A \vee x \in B\}$, $A \cap B = \{x \in U : x \in A \wedge x \in B\}$,

$B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$, $\overline{A} = \{x \in U : x \notin A\} = U \setminus A$.

Procedural versions (the steps in element proofs): $a \in X \cup Y \Leftrightarrow a \in X \vee a \in Y$; $a \in X \cap Y \Leftrightarrow a \in X \wedge a \in Y$; $a \in X \setminus Y \Leftrightarrow a \in X \wedge a \notin Y$; $a \in \overline{X} \Leftrightarrow a \notin X$; $(a, b) \in X \times Y \Leftrightarrow a \in X \wedge b \in Y$.

DEFINITION Symmetric difference (Tutorial 3) · Disjoint

$A \oplus B = (A \setminus B) \cup (B \setminus A)$, the set of elements in exactly one of A and B . A and B are **disjoint** iff $A \cap B = \emptyset$; $A_1, A_2, \dots$ are **mutually disjoint** iff $A_i \cap A_j = \emptyset$ whenever $i \neq j$. $\bigcup_{i=0}^n A_i$, $\bigcap_{i=0}^n A_i$ extend $\cup, \cap$.

THEOREM 6.2.1 · Subset relations

$A \cap B \subseteq A$, $A \cap B \subseteq B$ (inclusion of intersection); $A \subseteq A \cup B$, $B \subseteq A \cup B$ (inclusion in union); $A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$ (transitivity). The **set identities** (Theorem 6.2.2) are the right column of the table in §2.

PAST PAPER **AY23 Q7** · For which can no non-empty A, B exist?

(i) $A \setminus B = A \cap B$: these two sets are disjoint, so both must be $\emptyset$; then $A = (A \setminus B) \cup (A \cap B) = \emptyset$.

Impossible.

(ii) $A \setminus B = A \cup B$: then $B \subseteq A \cup B = A \setminus B$, so every element of B is not in B ; $B = \emptyset$. **Impossible.**

(iii) $A \cup B = A \cap B$: take $A = B$. (iv) $A \oplus B = A \cup B$: take $A \cap B = \emptyset$. Answer **B** (only (i), (ii)).

17 · Proving and disproving set statements

METHOD Three methods

Element method. To show $X \subseteq Y$: “Let $z \in X$... so $z \in Y$ ”. To show $X = Y$: prove both inclusions, or one chain of $\Leftrightarrow$ steps each justified by a definition or a logic law.

Algebraic method. Rewrite using the set identities (§2), one named law per line.

Disproof. Give one concrete counterexample with every set written out. A Venn diagram helps in finding one.

PROOF $\{x \in \mathbb{Z} : x^2 = 1\} = \{1, -1\}$ (two inclusions, Lecture 5)

1. ($\subseteq$) Take any $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
 - 1.1. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $(z-1)(z+1) = 0$ (by basic algebra).
 - 1.2. So $z = 1$ or $z = -1$, i.e. $z \in \{1, -1\}$.
2. ($\supseteq$) Take any $z \in \{1, -1\}$. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
3. Therefore the sets are equal (by definition of set equality, from 1 and 2). ■

PROOF De Morgan: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ (chain of $\Leftrightarrow$)

1. Let $z \in U$. Then
 - 1.1. $z \in \overline{A \cup B} \Leftrightarrow \sim(z \in A \cup B)$ (by definition of complement)
 - 1.2. $\Leftrightarrow \sim(z \in A \vee z \in B)$ (by definition of $\cup$)
 - 1.3. $\Leftrightarrow z \notin A \wedge z \notin B$ (by De Morgan's law for logic)
 - 1.4. $\Leftrightarrow z \in \overline{A} \wedge z \in \overline{B} \Leftrightarrow z \in \overline{A} \cap \overline{B}$ (by definitions of complement, $\cap$). ■

PROOF $(A \cap B) \cup (A \setminus B) = A$ (algebraic, Lecture 5)

$$\begin{aligned}
 (A \cap B) \cup (A \setminus B) &= (A \cap B) \cup (A \cap \overline{B}) && \text{(set difference law)} \\
 &= A \cap (B \cup \overline{B}) && \text{(distributive law)} \\
 &= A \cap U = A && \text{(complement law; identity law)}
 \end{aligned}$$

■

PROOF $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ (algebraic)

$$\begin{aligned}
 A \setminus (B \cap C) &= A \cap \overline{B \cap C} \quad \text{(set difference)} = A \cap (\overline{B} \cup \overline{C}) \quad \text{(De Morgan)} \\
 &= (A \cap \overline{B}) \cup (A \cap \overline{C}) \quad \text{(distributive)} = (A \setminus B) \cup (A \setminus C) \quad \text{(set difference } \times 2).
 \end{aligned}$$

■

PROOF For all sets A, B : $A \subseteq B \Leftrightarrow \mathcal{P}(A) \subseteq \mathcal{P}(B)$

1. ($\Rightarrow$) Suppose $A \subseteq B$. Let $X \in \mathcal{P}(A)$; then $X \subseteq A$ (definition of power set), so $X \subseteq B$ (transitivity of $\subseteq$), so $X \in \mathcal{P}(B)$.
2. ($\Leftarrow$) Suppose $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. $A \subseteq A$, so $A \in \mathcal{P}(A)$, hence $A \in \mathcal{P}(B)$, i.e. $A \subseteq B$ (definition of power set). ■

18 · Partitions and power sets

DEFINITION Partition (Lecture 6)

$\mathcal{C}$ is a **partition** of A iff (1) every element of $\mathcal{C}$ is a **non-empty** subset of A , and (2) every element of A lies in **exactly one** element of $\mathcal{C}$: $\forall x \in A \exists! S \in \mathcal{C} (x \in S)$. The elements of $\mathcal{C}$ are its **components**. (Equivalently: non-empty, mutually disjoint, union is A .)

DEFINITION Power set · Theorem 6.3.1

$\mathcal{P}(A)$ is the set of all subsets of A . If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$. Always $\emptyset \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$. $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$, but only $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

PAST PAPER AY24 Q7 and Q8 · Partitions, power sets

Q7. (i) $\{\{2, 3\}, \{4\}, \{3, 2\}, \{1\}\} = \{\{2, 3\}, \{4\}, \{1\}\}$ is a partition of $\{1, 2, 3, 4\}$, because repeated elements count once. (ii) $\{\mathbb{Z}\}$ is a partition of $\mathbb{Z}$ (one component). (iii) $\mathbb{Z}$ is **not**: its elements are numbers, not subsets. **C.**

Q8. (i) $|\mathcal{P}(S) \setminus \mathcal{P}(\emptyset)| = 2^{|S|} - 1$ is true, as $\mathcal{P}(\emptyset) = \{\emptyset\}$. (ii) $|S \cap \mathcal{P}(S)| = 0$ is false: $S = \{a, \{a\}\}$ gives $S \cap \mathcal{P}(S) = \{\{a\}\}$. (iii) $|S| \cdot |\mathcal{P}(S)| = |S \times \mathcal{P}(S)|$ is true. **D.**

PAST PAPER AY25 Q7, Q8 · Counting

Q7. A, B, C pairwise share exactly one element; $D = \mathcal{P}(A) \cap \mathcal{P}(B) \cap \mathcal{P}(C) = \mathcal{P}(A \cap B \cap C)$. The triple intersection lies inside each one-element pairwise intersection, so it has 0 or 1 elements: $|D| \in \{1, 2\}$. **B.**

Q8. $|A| = 2|B|$, $|B| = |C| + 2$, and $\{B, C, D\}$ partitions A , so $|A| = |B| + |C| + |D|$: $2|B| = |B| + (|B| - 2) + |D|$, giving $|D| = 2$. **C.**

PAST PAPER AY24 Q16 (5 marks) · $A = \{\emptyset, x\}$

$\mathcal{P}(A) = \{\emptyset, \{\emptyset\}, \{x\}, \{\emptyset, x\}\}$ and $\mathcal{P}(\emptyset) = \{\emptyset\}$, $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$.

- (a) $\mathcal{P}(A) \setminus \mathcal{P}(\mathcal{P}(\emptyset)) = \{\{x\}, \{\emptyset, x\}\}$.
- (b) $(\mathcal{P}(A) \setminus \mathcal{P}(\emptyset)) \setminus \{A\} = \{\{\emptyset\}, \{x\}\}$, so its power set is $\{\emptyset, \{\{\emptyset\}\}, \{\{x\}\}, \{\{\emptyset\}, \{x\}\}\}$.
- (c) $\mathcal{P}(\emptyset) \times \mathcal{P}(A) = \{(\emptyset, S) : S \in \mathcal{P}(A)\}$ and $\mathcal{P}(A) \times \mathcal{P}(\emptyset) = \{(S, \emptyset) : S \in \mathcal{P}(A)\}$ share only $(\emptyset, \emptyset)$, so the answer is $\{(\emptyset, \{\emptyset\}), (\emptyset, \{x\}), (\emptyset, \{\emptyset, x\})\}$.

PAST PAPER AY23 Q6 and Q17 · $A_1 = \{\emptyset\}$, $A_2 = \{a\}$, $A_3 = \{a, \{a\}\}$

Q6. Target $\emptyset \cup \mathcal{P}(\emptyset) = \{\emptyset\}$. $\emptyset \times \{\emptyset\} = \emptyset \narrow$; $\mathcal{P}(\emptyset \cup \mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\} \narrow$; $\mathcal{P}(\emptyset) \setminus \emptyset = \{\emptyset\} \checkmark$; $(\emptyset \cap \{\emptyset\}) \cup (\emptyset \cup \{\emptyset\}) = \{\emptyset\} \checkmark$. **D.**

Q17. (a) $(A_1 \cap A_2) \setminus A_3 = \emptyset$. (b) $(A_1 \cup A_2) \setminus A_3 = \{\emptyset\}$. (c) $(A_1 \times A_2) \cup A_3 = \{(\emptyset, a), a, \{a\}\}$. (d) $(A_1 \cup A_2) \times A_3 = \{(\emptyset, a), (\emptyset, \{a\}), (a, a), (a, \{a\})\}$.

More model proofs for sets

PROOF Distributive law $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, by the element method with cases (Epp 6.2)

1. ($\subseteq$) Let $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$ (definition of $\cup$).
 - 1.1. **Case $x \in A$:** then $x \in A \cup B$ and $x \in A \cup C$ (definition of $\cup$), so $x \in (A \cup B) \cap (A \cup C)$.
 - 1.2. **Case $x \in B \cap C$:** then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$.
2. ($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$, so $x \in A \cup B$ and $x \in A \cup C$.
 - 2.1. **Case $x \in A$:** then $x \in A \cup (B \cap C)$.
 - 2.2. **Case $x \notin A$:** then $x \in B$ and $x \in C$ (as $x \in A \cup B$, $x \in A \cup C$), so $x \in B \cap C \subseteq A \cup (B \cap C)$.
3. Hence the two sets are equal (definition of set equality). ■

PROOF For all sets A, B : $A \subseteq B \leftrightarrow A \cup B = B$

1. ($\rightarrow$) Suppose $A \subseteq B$. $B \subseteq A \cup B$ (inclusion in union). Conversely let $x \in A \cup B$: if $x \in A$ then $x \in B$ (as $A \subseteq B$) if $x \in B$, done. So $A \cup B = B$.
2. ($\leftarrow$) Suppose $A \cup B = B$. Let $x \in A$; then $x \in A \cup B$ (inclusion in union) $= B$. So $A \subseteq B$. ■

PROOF For every set A , $A \times \emptyset = \emptyset$ (proof by contradiction)

1. Suppose not: some $z \in A \times \emptyset$. Then $z = (a, b)$ with $a \in A$ and $b \in \emptyset$ (definition of $\times$).
2. But $\emptyset$ has no elements, a contradiction. So $A \times \emptyset$ has no elements: $A \times \emptyset = \emptyset$. ■

PROOF $A \times (B \cup C) = (A \times B) \cup (A \times C)$ (chain of $\leftrightarrow$)

$(x, y) \in A \times (B \cup C) \leftrightarrow x \in A \wedge (y \in B \vee y \in C)$ (definitions of $\times, \cup$) $\leftrightarrow (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$
 (distributive law of logic) $\leftrightarrow (x, y) \in (A \times B) \cup (A \times C)$ (definitions of $\times, \cup$). ■

PROOF $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$ (algebraic)

$(A \cup B) \setminus C = (A \cup B) \cap \overline{C}$ (set difference law) $= (A \cap \overline{C}) \cup (B \cap \overline{C})$ (distributive, commutative) $= (A \setminus C) \cup (B \setminus C)$
 (set difference law). ■

WORKED EXAMPLE Disproving a set identity (Epp 6.3) · “ $(A \setminus B) \cup (B \setminus C) = A \setminus C$ for all sets”
False. Counterexample: $A = \emptyset$, $B = \{1\}$, $C = \emptyset$. Then $(A \setminus B) \cup (B \setminus C) = \emptyset \cup \{1\} = \{1\}$, while $A \setminus C = \emptyset$.
 Write out every set and compute both sides.

DRILL Nested sets and power sets (answers after the arrow)

- $|\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}| \rightarrow 3$
- $\mathcal{P}(\{\emptyset\}) \rightarrow \{\emptyset, \{\emptyset\}\}$
- $\{\emptyset\} \times \{\emptyset\} \rightarrow \{(\emptyset, \emptyset)\}$
- $\emptyset \times \{\emptyset\} \rightarrow \emptyset$
- $\mathcal{P}(\{a, b\}) \setminus \{\emptyset\} \rightarrow \{\{a\}, \{b\}, \{a, b\}\}$
- $\{1, \{1\}\} \cap \mathcal{P}(\{1\}) \rightarrow \{\{1\}\}$
- $|\mathcal{P}(A \times B)|$ with $|A| = 2$, $|B| = 3 \rightarrow 2^6 = 64$
- $\{x \in \mathbb{Z} : x^2 < 5\} \rightarrow \{-2, -1, 0, 1, 2\}$
- $\{2n + 1 : n \in \{0, 1, 2\}\} \rightarrow \{1, 3, 5\}$
- $\{\{1, 2\}, \{3\}\}$ a partition of $\{1, 2, 3\}$? $\rightarrow$ **yes**;
 $\{\{1, 2\}, \{2, 3\}\} \rightarrow$ **no** (2 twice); $\{\{1, 2\}, \{3\}, \emptyset\} \rightarrow$
no ($\emptyset$ component)
- $A = \{\emptyset, \{\emptyset\}\}$: $A \cap \mathcal{P}(A) \rightarrow A$ itself, so $A \subseteq \mathcal{P}(A)$

PROOF The empty set is unique (Corollary 6.2.5)

1. Suppose $\emptyset_1$ and $\emptyset_2$ are both sets with no elements.
2. $\emptyset_1 \subseteq \emptyset_2$ and $\emptyset_2 \subseteq \emptyset_1$ (Theorem 6.2.4, applied to each as “the” empty set).
3. So $\emptyset_1 = \emptyset_2$ (definition of set equality). ■

PROOF If $A \subseteq B$ then $A \cap C \subseteq B \cap C$ and $A \cup C \subseteq B \cup C$

1. Let $x \in A \cap C$: $x \in A$ and $x \in C$ (definition of $\cap$) $x \in B$ (as $A \subseteq B$) so $x \in B \cap C$.
2. Let $x \in A \cup C$: if $x \in A$ then $x \in B \subseteq B \cup C$; if $x \in C$ then $x \in B \cup C$ (inclusion in union). ■

Reading only (Lecture 5.3). Unrestricted comprehension breaks: Russell’s $R = \{x : x \notin x\}$ gives $R \in R \leftrightarrow R \notin R$; a set of “everything” $\mathcal{D}$ contradicts $|\mathcal{P}(\mathcal{D})| > |\mathcal{D}|$ (Cantor). Zermelo (1908), later Fraenkel and Skolem, fixed this with the ZF axioms. Cantor, Russell, Zermelo and Fraenkel contributed to modern set theory; Gauss did not (AY25 Q1).

Part V · Relations

19 · Relations and inverses

DEFINITION Relation · Domain, co-domain, range

A (binary) **relation from A to B** is a subset $R \subseteq A \times B$; xRy means $(x, y) \in R$. A **relation on A** is a subset of $A \times A = A^2$.

$Dom(R) = \{a \in A : aRb \text{ for some } b \in B\}$, $coDom(R) = B$, $Range(R) = \{b \in B : aRb \text{ for some } a \in A\}$.

An **n -ary relation** on $A_1 \times \cdots \times A_n$ is a subset of it (binary, ternary, quaternary for $n = 2, 3, 4$).
 Relational databases are built on n -ary relations.

DEFINITION Inverse relation

$R^{-1} = \{(y, x) \in B \times A : (x, y) \in R\}$, i.e. $\forall x \in A, y \in B ((y, x) \in R^{-1} \leftrightarrow (x, y) \in R)$. Reverse every arrow.

PAST PAPER AY25 Q9, Q10 · Relations are sets of pairs

Q9. $A = \{1, 2, 3\}$, $B = \{2, 5\}$. $S_2 = \mathcal{P}(A \times B)$ is **exactly** the set S_3 of relations from A to B ; $S_1 = \mathcal{P}(A) \times \mathcal{P}(B)$ holds **pairs of sets**, a different kind of object. So $S_4 \subseteq S_3 \subseteq S_2$. Answer **A**.

Q10. The relation “ $p \rightarrow q$ ” on $\{T, F\}$ is $R = \{(T, T), (F, T), (F, F)\}$. $R^{-1} = \{(T, T), (T, F), (F, F)\}$ is “ $q \rightarrow p$ ”, the **converse**. The converse is equivalent to the inverse, so the answer key accepted **B or D**.

20 · Composition of relations (lecture-only; not in Epp)

DEFINITION Composition (Lecture 6.1.4)

For $R \subseteq A \times B$ and $S \subseteq B \times C$, the **composition** $S \circ R$ is the relation from A to C with

$$\forall x \in A, \forall z \in C (x(S \circ R)z \leftrightarrow \exists y \in B (xRy \wedge ySz)).$$

x and z are related iff there is a path $x \xrightarrow{R} y \xrightarrow{S} z$. $S \circ R$ applies R first.

THEOREM Associativity · Inverse of a composition

$$T \circ (S \circ R) = (T \circ S) \circ R = T \circ S \circ R. \quad (S \circ R)^{-1} = R^{-1} \circ S^{-1} \text{ (order reverses).}$$

Stated without proof in the lecture; the second follows by chasing the definition: $(z, x) \in (S \circ R)^{-1} \leftrightarrow \exists y (xRy \wedge ySz) \leftrightarrow \exists y (zS^{-1}y \wedge yR^{-1}x)$.

METHOD Computing $S \circ R$ by hand

For each pair $(x, y) \in R$, list every $(y, z) \in S$ that **starts where it ends**, and record (x, z) . For $R \circ R \circ R$, compute $R_2 = R \circ R$ first, then $R \circ R_2$. Stop and re-check pairs that could join through two different middles.

PAST PAPER AY25 Q17(a) (4 marks) · $S = \{(1, 3), (3, 6), (6, 9), (9, 12)\}$, $R = \{(6, 6), (9, 9)\}$

$R \circ S$ (apply S then R): $(3, 6)(6, 6) \rightarrow (3, 6)$; $(6, 9)(9, 9) \rightarrow (6, 9)$. So $R \circ S = \{(3, 6), (6, 9)\}$.

$S \circ R$ (apply R then S): $(6, 6)(6, 9) \rightarrow (6, 9)$; $(9, 9)(9, 12) \rightarrow (9, 12)$. So $S \circ R = \{(6, 9), (9, 12)\}$.

(i) $(R \circ S) \setminus (S \circ R) = \{(3, 6)\}$.

(ii) $(R^{-1} \circ S^{-1}) \circ (S \circ R) = (S \circ R)^{-1} \circ (S \circ R)$; with $(S \circ R)^{-1} = \{(9, 6), (12, 9)\}$ this is $\{(6, 6), (9, 9)\}$.

PAST PAPER AY24 Q11, Q17(e), Q12 · Powers and formula relations

Q11. $R = \{(a, a), (a, b), (b, c), (c, a), (c, b)\}$ on $\{a, b, c\}$. $R \circ R$ is all of $A \times A$ except (b, c) ; then bRc and $c(R \circ R)c$ put (b, c) into $R \circ R \circ R$, which is $A \times A$. Answer B.

Q17(e). $xR_1y \leftrightarrow y = 2x$, $xR_2y \leftrightarrow x = y^2$ on $\mathbb{Z}^+$. $x(R_1 \circ R_2)z \leftrightarrow \exists y (x = y^2 \wedge z = 2y)$, so $x = z^2/4$ (with z even).

Q12. $|R \circ R|$ has no fixed relation to $|R|$: on $\{a, b\}$, $R = \{(a, a), (a, b), (b, a)\}$ gives $|R \circ R| = 4 > 3$; $\{(a, a), (a, b)\}$ gives $2 = 2$ but $\neq 2^2$; $\{(a, a)\}$ gives $1 = 1^2$. None of (i)–(iii) always holds. Answer E.

PROOF $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$ (chain of $\leftrightarrow$)

For all $z \in C, x \in A$: $(z, x) \in (S \circ R)^{-1} \leftrightarrow (x, z) \in S \circ R$ (definition of inverse) $\leftrightarrow \exists y \in B (xRy \wedge ySz)$ (definition of $\circ$) $\leftrightarrow \exists y \in B (zS^{-1}y \wedge yR^{-1}x)$ (definition of inverse; commutative law) $\leftrightarrow (z, x) \in R^{-1} \circ S^{-1}$ (definition of $\circ$). ■

PROOF Composition is associative: $(T \circ S) \circ R = T \circ (S \circ R)$

$x((T \circ S) \circ R)w \leftrightarrow \exists y (xRy \wedge \exists z (ySz \wedge zTw)) \leftrightarrow \exists z \exists y (xRy \wedge ySz \wedge zTw)$ (z is not free in xRy ; same-kind quantifiers commute; associative law) $\leftrightarrow \exists z (\exists y (xRy \wedge ySz) \wedge zTw)$ (y is not free in zTw) $\leftrightarrow \exists z (x(S \circ R)z \wedge zTw) \leftrightarrow x(T \circ (S \circ R))w$. ■

21 · Properties of relations on a set A

DEFINITION Reflexive, symmetric, transitive (Lecture 6.2) · Antisymmetric (6.4) · Asymmetric (Tutorial 5)

	R has the property iff	R fails it iff
Reflexive	$\forall x \in A (xRx)$	$\exists x \in A (x \not R x)$
Symmetric	$\forall x, y \in A (xRy \rightarrow yRx)$	$\exists x, y (xRy \wedge y \not R x)$
Transitive	$\forall x, y, z \in A (xRy \wedge yRz \rightarrow xRz)$	$\exists x, y, z (xRy \wedge yRz \wedge x \not R z)$
Antisymmetric	$\forall x, y \in A (xRy \wedge yRx \rightarrow x = y)$	$\exists x, y (xRy \wedge yRx \wedge x \neq y)$
Asymmetric	$\forall x, y \in A (xRy \rightarrow y \not R x)$	$\exists x, y (xRy \wedge yRx)$

THEOREM Useful facts

- R symmetric $\Leftrightarrow R = R^{-1}$ (Tutorial 4).
- R transitive $\Leftrightarrow R \circ R \subseteq R$.
- R reflexive $\Rightarrow R \subseteq R \circ R$.
- Asymmetric $\Rightarrow$ antisymmetric (and irreflexive).
- **Antisymmetric is not “not symmetric”:** any $R \subseteq \{(x, x) : x \in A\}$ is both.
- “=” is an equivalence relation **and** a partial order (AY23 Q10, AY24 Q9).
- The empty relation on $A \neq \emptyset$ is symmetric, transitive, antisymmetric and asymmetric, but not reflexive.
- If R, S **both** have the property: reflexivity survives $\cup, \cap, \circ$; symmetry survives $\cup, \cap$ but not $\circ$; transitivity and antisymmetry survive only $\cap$.
- Write “0 is related to itself”, not “0 is reflexive”.

METHOD Proof skeletons for the properties

Reflexive: “Let $x \in A$ so xRx .” **Symmetric:** “Let $x, y \in A$ with xRy so yRx .”

Transitive: “Let $x, y, z \in A$ with xRy and yRz so xRz .” **Antisymmetric:** “Let xRy and yRx so $x = y$.”

Disproof: one explicit counterexample, named elements, each membership checked.

PAST PAPER AY23 Q9 and Q10 · Counterexamples on small sets

Q9 (R reflexive, S symmetric). On $\{a, b, c\}$ take $R = \{(a, a), (b, b), (c, c), (a, c)\}$, $S = \{(a, b), (b, a)\}$: $R \circ S = \{(a, b), (b, a), (b, c)\}$ is not symmetric; $R \cup S$ is not symmetric; $(R \circ S)^{-1}$ is not transitive. Only “ $R \cup S$ is reflexive” survives: **A**.

Q10. Asymmetric $\Rightarrow$ antisymmetric: true. “Not reflexive $\Rightarrow$ not asymmetric”: false ($R = \{(a, b)\}$). “Not reflexive $\Rightarrow$ asymmetric”: false ($R = \{(a, a)\}$ on $\{a, b\}$). “No relation is both an equivalence relation and a partial order”: false (=). Only (i): **A**.

PAST PAPER AY24 Q10, AY25 Q11, AY25 Q14

AY24 Q10. $pRq \leftrightarrow (p \rightarrow q)$ on $\{T, F\}$: $R = \{(T, T), (F, F), (F, T)\}$ is reflexive, antisymmetric, transitive, not symmetric: **D**.

AY25 Q11. $ARB \leftrightarrow |A \cap B| = 1$ on $\mathcal{P}(\mathbb{Z})$: symmetric; not reflexive ($|\emptyset \cap \emptyset| = 0$); not transitive. **Symmetric but not an equivalence relation.** Answer B.

AY25 Q14. Reflexive R on $\{1, 2, 3, 4\}$ with $(xRy \wedge xRz) \rightarrow y = z$: reflexivity forces xRx , so each x is related to **nothing else**. Only the identity relation qualifies, so $|\mathcal{F}| = 1$. Answer B.

PROOF R is symmetric $\Leftrightarrow R = R^{-1}$ (Tutorial 4)

1. ($\rightarrow$) Suppose R symmetric. $(x, y) \in R \rightarrow (y, x) \in R$ (symmetry) $\rightarrow (x, y) \in R^{-1}$ (definition of inverse), so $R \subseteq R^{-1}$. Conversely $(x, y) \in R^{-1} \rightarrow (y, x) \in R \rightarrow (x, y) \in R$ (symmetry), so $R^{-1} \subseteq R$.
2. ($\leftarrow$) Suppose $R = R^{-1}$ and xRy . Then $(y, x) \in R^{-1} = R$, i.e. yRx . ■

PROOF R is transitive $\Leftrightarrow R \circ R \subseteq R$ (AY23 Q12 model answer, extended)

1. ($\rightarrow$) Let $(x, z) \in R \circ R$. There is some y with xRy and yRz (definition of composition) so xRz (transitivity), i.e. $(x, z) \in R$.
2. ($\leftarrow$) Let xRy and yRz . Then $(x, z) \in R \circ R$ (definition of composition) $\subseteq R$, so xRz . ■

Similarly, **reflexive $\Rightarrow R \subseteq R \circ R$** : given $(x, z) \in R$, use xRx and xRz with middle $y = x$.

22 · Closures

DEFINITION Transitive closure (Lecture 6.2.2); reflexive and symmetric closures

The **transitive closure** of R on A is the relation R^t such that (1) R^t is transitive, (2) $R \subseteq R^t$, (3) $R^t \subseteq S$ for every transitive $S \supseteq R$. It is the **smallest** transitive relation containing R . Likewise the **reflexive closure** $R^r = R \cup \{(x, x) : x \in A\}$ and the **symmetric closure** $R^s = R \cup R^{-1}$.

METHOD Computing R^t

$R^t = R \cup R^2 \cup R^3 \cup \dots$ where $R^2 = R \circ R$; on a finite set stop once a power adds nothing new (at most $|A|$ powers). On the digraph: draw $x \rightarrow z$ whenever some path leads from x to z . A cycle back to the starting point creates a loop.

PAST PAPER AY24 Q17(a), (b) (5 marks)

(a) $R = \{(a, b), (a, c)\}$ on $\{a, b, c\}$: $|R^r| = 2 + 3 = 5$; $|R^s| = |\{(a, b), (a, c), (b, a), (c, a)\}| = 4$; $|R^t| = 2$, since no pair ends where another begins, so R is already transitive.

(b) $S = \{(a, b), (b, a), (b, d), (d, c), (b, b), (d, d)\}$; $T = S^t \setminus S$ consists of the new pairs obtained from paths: $a \rightarrow b \rightarrow a$ gives (a, a) ; $a \rightarrow b \rightarrow d$ gives (a, d) ; $a \rightarrow b \rightarrow d \rightarrow c$ gives (a, c) ; $b \rightarrow d \rightarrow c$ gives (b, c) . $T = \{(a, a), (a, c), (a, d), (b, c)\}$.

23 · Equivalence relations

DEFINITION Relation induced by a partition · Equivalence relation

Given a partition $\mathcal{C}$ of A , the **induced relation** is $xRy \leftrightarrow x, y$ lie in the same component; it is reflexive, symmetric and transitive (Theorem 8.3.1).

R on A is an **equivalence relation** iff it is **reflexive, symmetric and transitive**. Notation: $\sim$.

DEFINITION Equivalence class · Quotient $A/\sim$

$[a]_\sim = \{x \in A : a \sim x\}$. $A/\sim = \{[x]_\sim : x \in A\}$, “the quotient of A by $\sim$ ”. Its **elements** are the equivalence classes.

THEOREM Lemma Rel.1 · Theorem Rel.2 (Epp 8.3.4)

For an equivalence relation $\sim$ on A and $x, y \in A$ the following are equivalent: (i) $x \sim y$; (ii) $[x] = [y]$; (iii) $[x] \cap [y] \neq \emptyset$.

Hence $A/\sim$ is a **partition of A** , and equivalence relations on A correspond one-to-one with partitions of A .

The number of equivalence relations on an n -set is therefore the number of partitions (Bell numbers):

1, 2, 5, 15, 52 for $n = 1, \dots, 5$. For AY25 Q4 the answer is **5** on $\{1, 2, 3\}$.

PROOF Lemma Rel.1 (Lecture 6)

1. ((i) $\Rightarrow$ (ii)) Suppose $x \sim y$; then $y \sim x$ (*symmetry*).
 - 1.1. For every $z \in [x]$: $x \sim z$ (*definition of $[x]$*), so $y \sim z$ (*transitivity, as $y \sim x$*), so $z \in [y]$. Thus $[x] \subseteq [y]$.
 - 1.2. Symmetrically $[y] \subseteq [x]$, so $[x] = [y]$.
2. ((ii) $\Rightarrow$ (iii)) Suppose $[x] = [y]$. $x \sim x$ (*reflexivity*), so $x \in [x] = [x] \cap [y]$ (*idempotent law*) hence $[x] \cap [y] \neq \emptyset$.
3. ((iii) $\Rightarrow$ (i)) Take $z \in [x] \cap [y]$. Then $x \sim z$ and $y \sim z$ (*definitions of $[x], [y]$*) so $z \sim y$ (*symmetry*) and $x \sim y$ (*transitivity*). ■

PROOF Congruence mod n is an equivalence relation on $\mathbb{Z}$ ($a \equiv b \pmod{n} \leftrightarrow n \mid (a - b)$)

1. (Reflexive) For all $a \in \mathbb{Z}$: $a - a = 0 = n \cdot 0$, so $a \equiv a \pmod{n}$ (*definition of congruence*).
2. (Symmetric) Let $a \equiv b \pmod{n}$: $a - b = nk$ for some $k \in \mathbb{Z}$. Then $b - a = n(-k)$ with $-k \in \mathbb{Z}$ (*closure*), so $b \equiv a \pmod{n}$.
3. (Transitive) Let $a \equiv b$ and $b \equiv c \pmod{n}$: $a - b = nk$, $b - c = nl$. Then $a - c = n(k + l)$ with $k + l \in \mathbb{Z}$, so $a \equiv c \pmod{n}$. ■

PAST PAPER AY23 Q19 (6 marks) · $aSb \leftrightarrow \exists k \in \mathbb{Z} (a - b = k/2)$ **on** $\mathbb{Q}$

(a) **Prove S transitive** (official model answer):

1. Let $a, b, c \in \mathbb{Q}$ such that aSb and bSc .
2. There exist $m, n \in \mathbb{Z}$ such that $a - b = m/2$ and $b - c = n/2$ (by definition of S).
3. Now $a - c = (a - b) + (b - c) = m/2 + n/2 = (m + n)/2$ (by basic algebra).
4. $m + n \in \mathbb{Z}$ (by closure of integers under addition).
5. Therefore aSc (by definition of S). ■

(b) Two distinct classes: $[0] = \{k/2 : k \in \mathbb{Z}\}$ and $[0.1]$. (c) $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$. Each class has exactly one representative in $[0, 1/2)$, so there are no duplicates.

PAST PAPER AY25 Q17(c), (d), Q15 · AY24 Q9 · AY23 Q12

AY25 Q17(c). “Same biological father and mother”: **yes**, an equivalence relation. “Share a biological parent”: **no**, because it is not transitive (half-siblings).

AY25 Q17(d). $x \sim y \leftrightarrow \exists d \in \mathbb{Z}^+ (d \mid x \wedge d \mid y)$: $d = 1$ always works, so everything is related: $\mathbb{Z}^+ / \sim = \{\mathbb{Z}^+\} = \{[1]\}$.

AY25 Q15. “ $\forall x \in A ([x] \subseteq A/\sim)$ ” is **false**: classes are **elements** of $A/\sim$, not subsets. “ $\exists x \in A/\sim \forall y \in x (y \in A)$ ” and “ $|A| = \sum_{x \in A/\sim} |x|$ ” are true: **E**.

AY24 Q9. Divisibility on 10 primes is the identity relation: 10 classes **and** 10 minimal elements. Answer **C**.

AY23 Q12. For an equivalence relation R , all four hold: $R = R^{-1}$ (symmetry), so $R^{-1} \circ R = R \circ R^{-1} = R \circ R$; $R \circ R \subseteq R$ (transitivity: $xRy \wedge yRz \Rightarrow xRz$); $R \subseteq R \circ R$ (reflexivity: take the middle $y = x$). Hence $R \circ R^{-1} = R$: **D**.

PROOF Theorem Rel.2 · If $\sim$ is an equivalence relation on A , then $A/\sim$ is a partition of A (Lecture 6)

1. $A/\sim$ is a set (by definition).
2. Every element is a non-empty subset of A : $[x] = \{y \in A : x \sim y\} \subseteq A$, and $x \in [x]$ (reflexivity).
3. Every $x \in A$ is in exactly one class:
 - 3.1. $x \in [x]$, so x is in at least one class.
 - 3.2. If $x \in [y]$ and $x \in [z]$, then $[y] \cap [z] \neq \emptyset$, so $[y] = [z]$ (Lemma Rel.1, (iii) $\Rightarrow$ (ii)). ■

PROOF Theorem 8.3.1 · The relation induced by a partition $\mathcal{C}$ of A is an equivalence relation

1. (Reflexive) Let $x \in A$. x lies in some component S (definition of partition), so $x, x \in S$ and xRx .
2. (Symmetric) If xRy , then x, y lie in one component S , so $y, x \in S$ and yRx .
3. (Transitive) Let xRy and yRz : $x, y \in S_1$ and $y, z \in S_2$ for components S_1, S_2 . Then $y \in S_1 \cap S_2$, so $S_1 = S_2$ (each element lies in exactly one component), hence $x, z \in S_1$ and xRz . ■

PROOF Epp 8.3 · On $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, $(a, b) \sim (c, d) \leftrightarrow ad = bc$ is an equivalence relation

1. (Reflexive) $ab = ba$ (commutativity), so $(a, b) \sim (a, b)$.
2. (Symmetric) If $ad = bc$ then $cb = da$ (commutativity), so $(c, d) \sim (a, b)$.
3. (Transitive) Let $ad = bc$ and $cf = de$, with $b, d, f \neq 0$.
 - 3.1. Multiply: $adf = bcf = bde$, so $d(af - be) = 0$ (basic algebra).
 - 3.2. $d \neq 0$, so $af = be$ (zero product property), i.e. $(a, b) \sim (e, f)$. ■

Each class is one rational number: $[(1, 2)] = \{(1, 2), (2, 4), (-3, -6), \dots\}$; $A/\sim$ “is” $\mathbb{Q}$.

METHOD Writing $A/\sim$ without duplicates

Pick **one representative per class** from a region that meets every class exactly once, then write $\{[x] : x \in \text{that region}\}$.

- Congruence mod 3: $\mathbb{Z}/\equiv_3 = \{[0], [1], [2]\}$, with $[r] = \{3k + r : k \in \mathbb{Z}\}$.
- AY23 Q19: $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$.
- Lecture Ex. 15: nonempty $A \subseteq \{1, 2, 3\}$, related iff they have the same least element: $\{[1]\} = \{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$, $[2] = \{\{2\}, \{2, 3\}\}$, $[3] = \{\{3\}\}$, three classes.

24 · Partial orders

DEFINITION Partial order (Lecture 6.4.2) · Poset

R on A is a **partial order** iff it is **reflexive, antisymmetric and transitive**; then (A, R) is a **partially ordered set** (poset). A general partial order is written $\preceq$ (“curly less than or equal to”). Examples: $\leq$ on $\mathbb{R}$; $\subseteq$ on any set of sets; $|$ on $\mathbb{Z}^+$. Non-examples: $<$ (not reflexive); congruence mod n (not antisymmetric); $|$ on $\mathbb{Z}$ ($2 | -2$ and $-2 | 2$, but $2 \neq -2$).

PROOF $|$ is a partial order on any set $A \subseteq \mathbb{Z}^+$ (Lecture 6, Ex. 20)

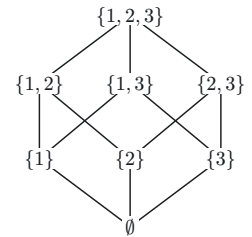
1. (Reflexive) Let $a \in A$. $a = 1 \cdot a$, so $a | a$ (by definition of divisibility).
2. (Antisymmetric) Let $a, b \in A$ with $a | b$ and $b | a$. Then $a \leq b$ and $b \leq a$ (by Theorem 4.4.1, as $a, b \in \mathbb{Z}^+$), so $a = b$.
3. (Transitive) Let $a | b$ and $b | c$. Then $a | c$ (by Theorem 4.4.3). ■

DEFINITION Hasse diagram (Lecture 6.4.3)

For a partial order $\preceq$ on a finite A and distinct x, y : if $x \preceq y$ and no $m \in A$ has $x \preceq m \preceq y$ (with $m \neq x, y$), draw x below y joined by a line; otherwise no line.

From the digraph: place arrows pointing up; delete (1) loops, (2) arrows implied by transitivity, (3) arrowheads.

Reading it: $x \preceq y$ iff you can climb from x to y along lines.



DEFINITION Comparable · Compatible (AY24 paper) · Chain, antichain (AY24 paper) · Total order

a, b are **comparable** iff $a \preceq b$ or $b \preceq a$; otherwise **noncomparable**. a, b are **compatible** iff $\exists c \in A$ ($a \preceq c \wedge b \preceq c$).

A **chain** is a subset C in which every two elements are comparable; a **maximal chain** M has no $t \notin M$ with $M \cup \{t\}$ a chain; the **length** of a chain is one less than its number of elements. An **antichain** is a subset in which no two distinct elements are comparable.

$\preceq$ is a **total (linear) order** iff it is a partial order and $\forall x, y \in A$ ($x \preceq y \vee y \preceq x$); its Hasse diagram is a single chain. $|$ on $\mathbb{Z}^+$ is partial, not total ($3 \nmid 5, 5 \nmid 3$); $\leq$ on $\mathbb{Q}$ is total.

DEFINITION Maximal, minimal, largest, smallest (Lecture 6.4.5)

For a poset $(A, \preceq)$ and $c \in A$:

- c **maximal** $\Leftrightarrow \forall x \in A$ ($c \preceq x \rightarrow c = x$): nothing lies strictly above c .
- c **minimal** $\Leftrightarrow \forall x \in A$ ($x \preceq c \rightarrow c = x$): nothing lies strictly below c .
- c **largest** $\Leftrightarrow \forall x \in A$ ($x \preceq c$): every element lies below c .
- c **smallest** $\Leftrightarrow \forall x \in A$ ($c \preceq x$): every element lies above c .

Largest = greatest = maximum; smallest = least = minimum. On a Hasse diagram: maximal = no line going up; minimal = no line going down.

THEOREM Extremal-element facts

- Smallest $\Rightarrow$ minimal; largest $\Rightarrow$ maximal.
- A largest (smallest) element, if it exists, is **unique**, and is then the **only** maximal (minimal) element.
- Conversely, in a **finite** poset a unique maximal element is largest. This fails for infinite posets.
- A finite non-empty poset has at least one maximal and one minimal element.
- Distinct maximal elements are noncomparable (so are distinct minimal ones).
- An element can be both maximal and minimal: an isolated point.
- Infinite posets may have none: $(\mathbb{Z}, \leq)$.
- **Well-ordered:** every non-empty subset has a smallest element. $(\mathbb{N}, \leq)$ is well-ordered; $(\mathbb{Z}, \leq)$ is not.

PROOF A smallest element is minimal (Lecture 6)

1. Let c be a smallest element, and take any $x \in A$ with $x \preceq c$.
2. By smallestness, $c \preceq x$ too. So $c = x$ (by antisymmetry). Hence c is minimal. ■

PROOF A largest element is unique

1. Suppose c and c' are both largest. Then $c' \preceq c$ (c largest) and $c \preceq c'$ (c' largest).
2. So $c = c'$ (by antisymmetry). ■

PAST PAPER AY23 Q13, Q14 · Facts about partial orders

(i) “No element is both smallest and largest” is false: $(\{a\}, \{(a, a)\})$. (ii) “An element can be maximal and minimal but neither largest nor smallest” is true: $(\{a, b\}, \{(a, a), (b, b)\})$. (iii) “Distinct maximal elements are comparable” is false, as it would contradict maximality. (iv) “There may be no maximal or minimal element” is true: $(\mathbb{Z}, \leq)$. **C.**

Q14. On $\{1, 2, 3\}$: $P = \{(1, 1), (1, 2), (2, 2)\}$ is not reflexive ($(3, 3)$ missing); $Q = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 2), (3, 3)\}$ is a partial order; $R = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 1)\}$ is not transitive ($1R2, 2R3$, not $1R3$), and $R^t = A \times A$ is not antisymmetric. Only Q : **B.**

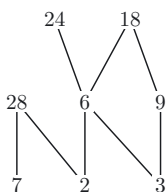
25 · Linearization and Kahn’s algorithm (lecture-only; Epp calls it topological sorting)**DEFINITION Linearization (Lecture 6.4.7)**

A **linearization** of a partial order $\preceq$ on A is a **total order** $\preceq^*$ on A with $\forall x, y \in A (x \preceq y \rightarrow x \preceq^* y)$. It lists the elements one at a time without breaking any precedence.

METHOD Kahn’s algorithm (1962) · Testing a proposed line-up

Kahn: set $A_0 := A$, $i := 0$; while $A_i \neq \emptyset$: pick a **minimal** element c_i of A_i , set $A_{i+1} := A_i \setminus \{c_i\}$, $i := i + 1$. Output $c_0 \preceq^* c_1 \preceq^* \dots$. Different choices of minimal element give different linearizations.

Test: a sequence is a linearization iff **for every line of the Hasse diagram, the lower element comes earlier**. A single reversed pair means it is not a linearization.

**PAST PAPER AY24 Q13–Q15 · $(\{2, 3, 6, 7, 9, 18, 24, 28\}, |)$**

Covers: $2 \mid 6, 2 \mid 28, 3 \mid 6, 3 \mid 9, 7 \mid 28, 6 \mid 18, 6 \mid 24, 9 \mid 18$.

Q13 Longest maximal chains have length 2:

$\{2, 6, 18\}, \{2, 6, 24\}, \{3, 6, 18\}, \{3, 6, 24\}, \{3, 9, 18\}$. There are **5**; answer **B**.

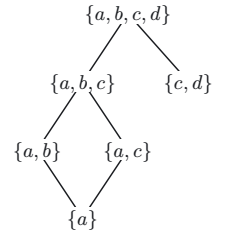
Q14 $\{2, 3, 28\}$ is not an antichain ($2 \mid 28$); $\{7, 9, 24\}$ is; $\{6\}$ is, vacuously. **D.**

Q15 $2, 3, 6, 7, 9, 18, 24, 28$ ✓; $7, 3, 2, 28, 9, 6, 18, 24$ ✓; $7, 2, 3, 6, 18, 9, 24, 28$ ✗, since $9 \mid 18$ but 18 comes first. **B.**

PAST PAPER AY24 Q17(c), (d) · $(E = \{2, 4, 6, 8, 10, 12\}, |)$

Covers: $2 \mid 4, 2 \mid 6, 2 \mid 10, 4 \mid 8, 4 \mid 12, 6 \mid 12$. **Minimal:** 2. **Maximal:** 8, 10, 12. **Smallest:** 2. **Largest:** none. Two maximal chains of different lengths: $\{2, 4, 8\}$ (length 2) and $\{2, 10\}$ (length 1); $\{2, 4, 12\}, \{2, 6, 12\}$ also work for the first.

PAST PAPER AY25 Q17(e) · $(\{\{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, \{a, b, c, d\}, \{c, d\}\}, \subseteq)$
 Covers: $\{a\} \subset \{a, b\}, \{a, c\} \subset \{a, b, c\} \subset \{a, b, c, d\}$, and $\{c, d\} \subset \{a, b, c, d\}$. **Smallest element:** no, since $\{a\}$ and $\{c, d\}$ are both minimal and noncomparable. **Maximal:** yes, $\{a, b, c, d\}$ (which is also largest).



PAST PAPER AY23 Q15, Q18 · $x \preceq z \leftrightarrow F_x = F_z \vee |F_x| < |F_z|$ on $\{11, \dots, 16\}$, $F_x =$ positive divisors
 Divisor counts: $11 \rightarrow 2, 13 \rightarrow 2, 14 \rightarrow 4, 15 \rightarrow 4, 16 \rightarrow 5, 12 \rightarrow 6$. The order compares counts only, so 11, 13 are incomparable, as are 14, 15. **Minimal:** 11, 13. **Smallest:** none. **Maximal:** 12. **Largest:** 12.
Q15: a linearization must list counts in non-decreasing order: 13, 11, 15, 14, 16, 12 ✓ and 11, 13, 14, 15, 16, 12 ✓; the other two put 12 too early. **B.**

PROOF AY24 Q17(f), (g) · $(a, b)W(c, d) \leftrightarrow a \leq c \wedge b \geq d$ on $\mathbb{Z} \times \mathbb{Z}$
(f) “Every two elements are comparable” is false. Counterexample: $(1, 3)$ and $(2, 4)$, since neither $1 \leq 2 \wedge 3 \geq 4$ nor $2 \leq 1 \wedge 4 \geq 3$ holds.
(g) “Every two elements are compatible” is true.
 1. Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be arbitrary elements of $\mathbb{Z} \times \mathbb{Z}$.
 2. Let $c = (\max(x_1, y_1), \min(x_2, y_2)) \in \mathbb{Z} \times \mathbb{Z}$.
 3. Then $x_1 \leq c_1, y_1 \leq c_1$ (definition of max) and $x_2 \geq c_2, y_2 \geq c_2$ (definition of min).
 4. So xWc and yWc (definition of W): x and y are compatible. ■

Relation	Refl.	Sym.	Trans.	Antisym.	Verdict
$=$ on any set	✓	✓	✓	✓	equivalence and partial order
$\leq$ on $\mathbb{R}$	✓	✗	✓	✓	total order
$<$ on $\mathbb{R}$	✗	✗	✓	✓	neither (asymmetric)
$ $ on $\mathbb{Z}^+$	✓	✗	✓	✓	partial order, not total
$ $ on $\mathbb{Z}$	✓	✗	✓	✗	neither ($2 \mid -2 \mid 2$)
$\equiv \pmod{n}$ on $\mathbb{Z}$	✓	✓	✓	✗	equivalence, n classes
$\subseteq$ on $\mathcal{P}(A)$	✓	✗	✓	✓	partial order
$ X = Y $ on $\mathcal{P}(A)$	✓	✓	✓	✗	equivalence, classes by size
$xy \geq 0$ on $\mathbb{Z}$	✓	✓	✗	✗	neither: $-1, 0, 1$
$ x - y \leq 1$ on $\mathbb{Z}$	✓	✓	✗	✗	neither: $0, 1, 2$
$\emptyset$ on $A \neq \emptyset$	✗	✓	✓	✓	neither
$A \times A$	✓	✓	✓	✗ if $ A \geq 2$	equivalence, one class

WORKED EXAMPLE Lecture 6, Ex. 26 · Kahn’s algorithm on $\{d \in \mathbb{Z}^+ : d \mid 30\}$

Step	Minimal elements of A_i	Choose
A_0	1	$c_0 = 1$
A_1	2, 3, 5	$c_1 = 3$
A_2	2, 5	$c_2 = 2$
A_3	5, 6	$c_3 = 6$
A_4	5	$c_4 = 5$
A_5	10, 15	$c_5 = 15$
A_6	10	$c_6 = 10$
A_7	30	$c_7 = 30$

Result: $1 \preceq^* 3 \preceq^* 2 \preceq^* 6 \preceq^* 5 \preceq^* 15 \preceq^* 10 \preceq^* 30$. Any other choice at steps 1–3 or 5 gives another valid linearization.

WORKED EXAMPLE Lecture 6 · Composition as “going through” a middle set

“takes” $\subseteq$ Students $\times$ Modules and “held in” $\subseteq$ Modules $\times$ Venues. Ann takes CS1010, CS1231, MA1101; Bryan takes CS1010, IS1103; Candy takes CS1231, IS1103; Danny takes nothing. CS1010 is held in LT15; CS1231 in ICube and SR1; IS1103 in SR1; CS2100 in LT15; MA1101 nowhere.

“held in $\circ$ takes” = “goes to”: Ann $\rightarrow$ {LT15, ICube, SR1}; Bryan $\rightarrow$ {LT15, SR1}; Candy $\rightarrow$ {ICube, SR1}; Danny $\rightarrow$ nothing. CS2100 contributes no pair, since nobody takes it.

THEOREM Closure facts

The transitive closure of a **symmetric** relation is symmetric, and of a **reflexive** relation is reflexive (reverse or extend the paths). Hence the **smallest equivalence relation containing R** is $(R \cup R^{-1} \cup \{(x, x) : x \in A\})^t$. Its classes are the connected components of the digraph of R when arrow directions are ignored.

Part VI · Practice with Model Solutions

New problems in the style of Part B. Attempt each one before reading the solution, and compare the justifications line by line.

PRACTICE 1 · Logic laws

Without a truth table, show $(p \rightarrow q) \wedge (p \rightarrow \sim q) \equiv \sim p$.

PROOF Solution

$(p \rightarrow q) \wedge (p \rightarrow \sim q) \equiv (\sim p \vee q) \wedge (\sim p \vee \sim q)$ (implication law $\times 2$) $\equiv \sim p \vee (q \wedge \sim q)$ (distributive law) $\equiv \sim p \vee \mathbf{c}$ (negation law) $\equiv \sim p$ (identity law). ■

PRACTICE 2 · Validity

Decide validity: (a) $p \rightarrow q, q \rightarrow r, \sim r \therefore \sim p$. (b) $p \rightarrow q, \sim p \therefore \sim q$.

PROOF Solution

(a) **Valid:** $p \rightarrow r$ (transitivity) with $\sim r, \sim p$ (modus tollens).
 (b) **Invalid** (inverse error): the critical row $p = F, q = T$ makes both premises true and the conclusion $\sim q$ false.

PRACTICE 3 · Nested quantifiers

Decide and justify: (a) $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (y > x^2)$; (b) $\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (y > x^2)$.

PROOF Solution

(a) **True.** Let $x \in \mathbb{Z}$ be arbitrary; take $y = x^2 + 1 \in \mathbb{Z}$ (closure) then $y > x^2$.
 (b) **False.** Its negation $\forall y \in \mathbb{Z} \exists x \in \mathbb{Z} (y \leq x^2)$ holds: given y , take $x = y$; then $y \leq y^2$ for every integer y (if $y \leq 0$ then $y \leq 0 \leq y^2$; if $y \geq 1$ then $y^2 = y \cdot y \geq y$).

PRACTICE 4 · Division into cases

Prove: for every integer n , $n^2 + n$ is even.

PROOF Solution

- Let n be a particular but arbitrarily chosen integer. Then n is even or odd (Assumption 1).
 - Case 1:** $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, and $2k^2 + k \in \mathbb{Z}$ (closure).
 - Case 2:** $n = 2k + 1$. Then $n^2 + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$, and $(2k + 1)(k + 1) \in \mathbb{Z}$ (closure).
- In both cases $n^2 + n$ is even (by definition of even). ■

PRACTICE 5 · Contradiction

Prove: there is no smallest positive rational number.

PROOF Solution

- Suppose not: let r be the smallest positive rational number.
 - $r = a/b$ for some integers a, b with $b \neq 0$ (definition of rational). Then $r/2 = a/(2b)$, where $a, 2b \in \mathbb{Z}$ and $2b \neq 0$, so $r/2$ is rational.
 - $r/2 > 0$ and $r/2 < r$ (as $r > 0$), contradicting the choice of r as the smallest.
- Hence there is no smallest positive rational number. ■

PRACTICE 6 · Contraposition

Prove: for all integers a, b , if ab is even then a is even or b is even.

PROOF Solution

1. Contrapositive: for all integers a, b , if a is odd and b is odd, then ab is odd.
2. Let a, b be odd: $a = 2r + 1$, $b = 2s + 1$ for some $r, s \in \mathbb{Z}$ (definition of odd).
 - 2.1. $ab = 4rs + 2r + 2s + 1 = 2(2rs + r + s) + 1$ (basic algebra), with $2rs + r + s \in \mathbb{Z}$ (closure) so ab is odd.
3. Hence the original statement holds (a statement is equivalent to its contrapositive). ■

PRACTICE 7 · Set identity

Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

PROOF Solution

$$\begin{aligned} A \setminus (B \cup C) &= A \cap \overline{B \cup C} \quad (\text{set difference law}) = A \cap (\overline{B} \cap \overline{C}) \quad (\text{De Morgan's law}) = (A \cap A) \cap (\overline{B} \cap \overline{C}) \quad (\text{idempotent law}) \\ &= (A \cap \overline{B}) \cap (A \cap \overline{C}) \quad (\text{associative, commutative laws}) = (A \setminus B) \cap (A \setminus C) \quad (\text{set difference law}). \quad \blacksquare \end{aligned}$$

PRACTICE 8 · Prove or disprove

(a) $A \cup (B \setminus C) = (A \cup B) \setminus C$; (b) $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$; (c) $A \subseteq C \wedge B \subseteq C \rightarrow A \cup B \subseteq C$.

PROOF Solution

- (a) **False:** $A = \{1\}$, $B = \emptyset$, $C = \{1\}$: left side $\{1\}$, right side $\emptyset$.
- (b) **False:** $A = \{1\}$, $B = \{2\}$: $\{1, 2\} \in \mathcal{P}(A \cup B)$ but $\{1, 2\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$. (Only $\supseteq$ holds.)
- (c) **True:** let $x \in A \cup B$; then $x \in A$ or $x \in B$ (definition of $\cup$) in either case $x \in C$ (as $A \subseteq C$, $B \subseteq C$). ■

PRACTICE 9 · Equivalence relation and quotient

On $\mathbb{Z}$ let $xRy \leftrightarrow 3 \mid (x^2 - y^2)$. Prove R is an equivalence relation and find $\mathbb{Z}/R$.

PROOF Solution

1. (Reflexive) $x^2 - x^2 = 0 = 3 \cdot 0$, so xRx (definition of divisibility).
2. (Symmetric) If $x^2 - y^2 = 3k$ then $y^2 - x^2 = 3(-k)$ with $-k \in \mathbb{Z}$, so yRx .
3. (Transitive) If $x^2 - y^2 = 3k$ and $y^2 - z^2 = 3l$, then $x^2 - z^2 = 3(k + l)$, so xRz .
4. (Classes) By the QR theorem $x = 3q + r$ with $r \in \{0, 1, 2\}$; $x^2 = 9q^2 + 6qr + r^2$, so $x^2 \bmod 3 = 0$ if $r = 0$ and 1 if $r = 1, 2$. Hence $\mathbb{Z}/R = \{[0], [1]\} = \{\{3k : k \in \mathbb{Z}\}, \{n \in \mathbb{Z} : 3 \nmid n\}\}$. ■

PRACTICE 10 · Not an equivalence relation

On $\mathbb{Z}$ let $xRy \leftrightarrow xy \geq 0$. Which properties hold?

PROOF Solution

Reflexive ($x^2 \geq 0$) and **symmetric** ($xy = yx$). **Not transitive:** $(-1)R0$ and $0R1$ (both products are 0), but $(-1)(1) = -1 < 0$. So R is not an equivalence relation. (On $\mathbb{Z} \setminus \{0\}$ it is an equivalence relation, “has the same sign as”, with classes $\mathbb{Z}^+$ and $\mathbb{Z}^-$.)

PRACTICE 11 · Partial order and extremal elements

On $\mathbb{Z}^+$ let $a \preccurlyeq b \leftrightarrow b = a \cdot 2^k$ for some $k \in \mathbb{N}$. Prove $\preccurlyeq$ is a partial order; find its minimal and maximal elements.

PROOF Solution

1. (Reflexive) $a = a \cdot 2^0$ and $0 \in \mathbb{N}$.
2. (Antisymmetric) If $b = a2^j$ and $a = b2^k$, then $a = a2^{j+k}$, so $2^{j+k} = 1$, $j = k = 0$, and $a = b$.
3. (Transitive) If $b = a2^j$ and $c = b2^k$ then $c = a2^{j+k}$ with $j + k \in \mathbb{N}$ (closure).
4. (Minimal) $b \preccurlyeq a$ with $b \neq a$ means $a = b2^k$, $k \geq 1$, which is possible iff a is even. So the minimal elements are exactly the **odd** numbers.
5. (Maximal) None: $a \preccurlyeq 2a$ and $2a \neq a$. Hence no largest or smallest element either. ■

PRACTICE 12 · Composition and closure

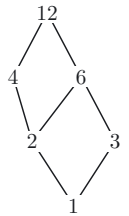
$A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Find $R \circ R$, $R^{-1} \circ R$ and $|R^t|$.

PROOF Solution

$R \circ R = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$ (two steps round the 4-cycle). $R^{-1} \circ R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$: follow an arrow of R , then return along the same arrow. From any vertex a path reaches every vertex, itself included, so $R^t = A \times A$ and $|R^t| = 16$.

PRACTICE 13 · Hasse diagram and Kahn

For $(\{1, 2, 3, 4, 6, 12\}, |)$: draw the Hasse diagram; give the minimal, maximal, smallest, largest elements; produce a linearization with Kahn's algorithm.



PROOF Solution

Covers: $1 \mid 2, 1 \mid 3, 2 \mid 4, 2 \mid 6, 3 \mid 6, 4 \mid 12, 6 \mid 12$ (diagram right). Minimal and smallest: **1**.

Maximal and largest: **12**. Kahn: A_0 has the single minimal element 1; then 2 and 3 are minimal and we take 2; then 3 and 4, and we take 3; then 4 and 6, and we take 4; then 6; then 12.

Linearization: $1 \prec^* 2 \prec^* 3 \prec^* 4 \prec^* 6 \prec^* 12$ (also valid: $1, 3, 2, 6, 4, 12$).

PRACTICE 14 · Power set of a set that contains $\emptyset$

Let $A = \{\emptyset, \{\emptyset\}\}$. Find $\mathcal{P}(A)$ and $A \cap \mathcal{P}(A)$.

PROOF Solution

$\mathcal{P}(A) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$, with four subsets since $|A| = 2$. For $A \cap \mathcal{P}(A)$ test each element of A : $\emptyset \in \mathcal{P}(A)$ (as $\emptyset \subseteq A$), and $\{\emptyset\} \in \mathcal{P}(A)$ (as $\{\emptyset\} \subseteq A$, since $\emptyset \in A$). So $A \cap \mathcal{P}(A) = \{\emptyset, \{\emptyset\}\} = A$, i.e. $A \subseteq \mathcal{P}(A)$.

PRACTICE 15 · Classes of a relation on pairs

On $\mathbb{Z} \times \mathbb{Z}$ let $(a, b) \sim (c, d) \leftrightarrow a + d = b + c$. Show $\sim$ is an equivalence relation and describe $(\mathbb{Z} \times \mathbb{Z})/\sim$.

PROOF Solution

1. Rewrite: $(a, b) \sim (c, d) \leftrightarrow a - b = c - d$ (*basic algebra*). Then reflexivity, symmetry and transitivity are those of $=$ on $\mathbb{Z}$ (e.g. $a - b = c - d$ and $c - d = e - f$ give $a - b = e - f$).
2. So $[(a, b)] = \{(c, d) : c - d = a - b\}$, a diagonal line of lattice points, and each class contains exactly one point $(k, 0)$ with $k = a - b$.
3. $(\mathbb{Z} \times \mathbb{Z})/\sim = \{[(k, 0)] : k \in \mathbb{Z}\}$, with one class for each integer and no duplicates. ■

PRACTICE 16 · All the closures

$A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 3)\}$. Find R^r , R^s , R^t and the smallest equivalence relation containing R .

PROOF Solution

$R^r = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$ (5 pairs); $R^s = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ (4); $R^t = \{(1, 2), (2, 3), (1, 3)\}$ (3).

The smallest equivalence relation containing R is $A \times A$ (9 pairs): ignoring directions, 1–2–3 is one connected piece, so there is a single class.

Part VII · Exam Playbook and Definition Index

26 · Part A: multiple-choice questions

METHOD Approach

- **Judge each of (i)–(iv) on its own** and mark T/F in the margin before reading the options. Several options usually differ in only one item. “None of (A)–(D)” has been the correct answer several times (AY23 Q2, AY24 Q12, AY25 Q2).
- **“For all sets / all relations” claims:** look for a counterexample among $\emptyset$, a singleton, $\{a, b\}$, the empty relation and the identity relation. One counterexample is enough.
- **Relations on small sets:** draw the digraph. **Posets:** draw the Hasse diagram first, then read off the answers.
- **Nested $\emptyset$ / power sets:** write each layer out explicitly, especially for $\mathcal{P}$ of a set containing $\emptyset$.
- **Counting questions:** use the table in §28, or enumerate the $n = 2$ case to test an option.

Common error	Correct statement	Paper
$\forall x(P \wedge Q)$ read as “all P are Q ”	It says everything is P and Q ; “all P are Q ” is $\forall x(P \rightarrow Q)$	AY23 Q2, AY24 Q2
A $\forall$ -statement needs a non-empty domain	Over $\emptyset$ it is vacuously true	AY23 Q2, AY25 Q2
Swapping $\forall$ and $\exists$	$\forall x \exists y$ lets y depend on x ; $\exists y \forall x$ does not	AY23 Q5
$\emptyset = \{\emptyset\}$, or $\mathcal{P}(\emptyset) = \emptyset$	$\mathcal{P}(\emptyset) = \{\emptyset\}$ has one element	AY23 Q6, AY24 Q16
$\in$ is transitive	Only $\subseteq$ is: $A \in B \subseteq C \Rightarrow A \in C$, but $A \subseteq B \in C$ gives nothing	AY23 Q8
A partition may repeat components or be a set of numbers	Duplicates collapse; components must be non-empty subsets	AY24 Q7
$A \times B \subseteq B \times C \Rightarrow A \subseteq C$	Fails when $B = \emptyset$	AY24 Q6
$S \circ R$ applies S first	R first: xRy , then ySz	AY24 Q17, AY25 Q17
Antisymmetric = not symmetric	Independent; subsets of the identity are both	AY23 Q10
Equivalence relations and partial orders are disjoint	$=$ is both; so is divisibility on primes	AY23 Q10, AY24 Q9
$[x] \subseteq A/\sim$	$[x] \in A/\sim$ and $[x] \subseteq A$	AY25 Q15
Maximal = largest	Largest needs everything below it; several maximal elements rule it out	AY23 Q13, Q18
Distinct maximal elements may be comparable	Never; that contradicts maximality	AY23 Q13
Any ordering of the elements is a linearization	Every Hasse edge must point forward in the line-up	AY23 Q15, AY24 Q15
$ R \circ R $ relates simply to $ R $	It can be larger, equal or smaller	AY24 Q12

27 · Part B: written answers

METHOD Computations

- Answer in **set-roster notation** with braces; list every element; working is usually not required.
- Write **“None”** rather than leaving a blank; a blank is treated as no answer (AY23 Q18, AY24 Q17c).
- For $A/\sim$, use one representative per class, with **no duplicates** (AY23 Q19c).
- Keep answers short; **marks may be deducted for excessively long answers.**

METHOD Proofs

- State the goal, then **unpack every definition** you use.
- One claim per numbered line, each with its justification.
- For “prove or disprove”, **state which first**, then the proof or one explicit counterexample.
- Finish by restating what was shown.

28 · Counting facts

Relations on an n -element set	General	$n = 2$	$n = 3$
all relations	2^{n^2}	16	512
reflexive	2^{n^2-n}	4	64
symmetric	$2^{n(n+1)/2}$	8	64
reflexive and symmetric	$2^{n(n-1)/2}$	2	8
antisymmetric	$2^n \cdot 3^{n(n-1)/2}$	12	216
asymmetric	$3^{n(n-1)/2}$	3	27
equivalence relations (Bell)	B_n	2	5
partial orders	—	3	19
total orders	$n!$	2	6

Derivation. Each of the n^2 pairs is either in the relation or not; reflexivity fixes the n diagonal pairs; symmetry decides each unordered pair once; antisymmetry allows 3 of the 4 states of each off-diagonal pair; asymmetry also forbids loops. For $n = 4$ there are 15 equivalence relations, 219 partial orders and 24 total orders. **AY25 Q12:** on $\{0, 1\}$ there are 2 equivalence relations and 3 partial orders, so $|F| + 1 = |G|$. Answer B.

Other counts	Value
$ A \times B $	$ A \cdot B $
$ \mathcal{P}(A) $	$2^{ A }$
relations from A to B	$2^{ A \cdot B }$
partitions of an n -set	$B_n : 1, 2, 5, 15, 52$

29 · Proof skeletons and past-paper locator

PROOF $\forall x \in D (P(x) \rightarrow Q(x))$, **direct**

- Let x be a particular but arbitrarily chosen element of D with $P(x)$.
 - Unpack $P(x)$ (by definition of ...).
 - Algebra / earlier results (by ...).
 - Repack as $Q(x)$ (by definition of ...).
- Therefore $\forall x \in D (P(x) \rightarrow Q(x))$. ■

PROOF **By contradiction**

- Suppose not: $\sim S$ (write the negation out explicitly).
 - Derive consequences, each justified.
 - Reach R and $\sim R$ for some statement R .
- Hence the supposition is false, so S . ■

PROOF **By contraposition**

- Contrapositive: $\forall x \in D (\sim Q(x) \rightarrow \sim P(x))$.
- Let $x \in D$ be arbitrary with $\sim Q(x)$ so $\sim P(x)$.
- Hence the original statement holds (equivalent to its contrapositive). ■

PROOF **Division into cases**

- Let x be arbitrary. Then x satisfies case 1, 2, ... (Assumption 1 / QR theorem / trichotomy).
 - Case 1:** ... so $Q(x)$.
 - Case 2:** ... so $Q(x)$.
- In every case $Q(x)$. ■

PROOF **Set inclusion $X \subseteq Y$ and equality**

- $(\subseteq)$ Let $z \in X$ so $z \in Y$.
- $(\supseteq)$ Let $z \in Y$ so $z \in X$.
- Therefore $X = Y$ (definition of set equality). ■

PROOF **R is an equivalence relation / a partial order**

- (Reflexive) Let $x \in A$ xRx .
- (Symmetric) Let xRy yRx . **or**
(Antisymmetric) Let xRy and yRx $x = y$.
- (Transitive) Let xRy and yRz xRz .
- Therefore R is an equivalence relation / a partial order. ■

Past-paper question	Solved in
AY23 Q2, Q4, Q5, Q16	§6, §4, §7, §7
AY23 Q3, Q6, Q7, Q8, Q17	§2, §18, §16, §14, §18
AY23 Q9, Q10, Q12, Q13, Q14, Q15, Q18, Q19	§21, §21, §23, §24, §24, §25, §25, §23
AY24 Q2, Q3, Q4, Q5	§5, §2, §2, §4
AY24 Q6, Q7, Q8, Q16	§15, §18, §18, §18
AY24 Q9, Q10, Q11, Q12, Q13–15, Q17	§23, §21, §20, §20, §25, §20–§25
AY25 Q1, Q2, Q4, Q6, Q7, Q8, Q9, Q10, Q12	§18, §6, §23, §8, §18, §18, §19, §19, §28
AY25 Q11, Q13, Q14, Q15, Q16, Q17	§21, §4, §21, §23, §4; Q17(a)–(e): §20, §13, §23, §23, §25

30 · Self-test: true or false?

Decide each statement before reading the reason.

Statement	Reason
-----------	--------

$\sim(p \leftrightarrow q) \equiv \sim p \leftrightarrow q$	T	both are true exactly when p, q differ
$(p \vee q) \rightarrow r \equiv (p \rightarrow r) \vee (q \rightarrow r)$	F	it is $\wedge$; $p = T, q = F, r = F$ separates them
$\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (x < y)$	F	take $x = y$
$\exists x \in \emptyset P(x)$	F	no witness exists; $\forall x \in \emptyset P(x)$ is T
$\emptyset \in \{\emptyset\}$ and $\emptyset \subseteq \{\emptyset\}$	T	listed element; Theorem 6.2.4
$\{\emptyset\} \subseteq \emptyset$	F	$\emptyset \in \{\emptyset\}$ but $\emptyset \notin \emptyset$
$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$	T	$X \subseteq A \cap B \leftrightarrow X \subseteq A \wedge X \subseteq B$
$A \times B = B \times A \Rightarrow A = B$	F	$A = \emptyset, B = \{1\}$: both products are $\emptyset$
a partition of a finite A has at most $ A $ components	T	components are non-empty and disjoint
symmetric and transitive $\Rightarrow$ reflexive	F	the empty relation on $\{a\}$
$R \circ R^{-1}$ is always symmetric	T	$(R \circ R^{-1})^{-1} = (R^{-1})^{-1} \circ R^{-1}$
the inverse of a partial order is a partial order	T	each property survives reversing pairs
the union of two equivalence relations is one	F	$1 \sim 2$ and $2 \sim 3$ from different relations
the intersection of two equivalence relations is one	T	each property is preserved by $\cap$
a unique minimal element is smallest, in any poset	F	true only for finite posets (§24)
$(\mathbb{Z}^+,)$ has a smallest element	T	$1 \mid n$ for all n
there are 4 relations on a one-element set	F	$2^{1^2} = 2$: $\emptyset$ and $\{(a, a)\}$
$a \mid b \wedge a \mid c \Rightarrow a \mid (b - c)$	T	$b - c = a(r - s)$

31 · Symbol table

Symbol	Read as	Symbol	Read as	Symbol	Read as
$\sim p$	not p	$p \wedge q$	p and q	$p \vee q$	p or q
$p \rightarrow q$	if p then q	$p \leftrightarrow q$	p iff q	$p \oplus q$	exclusive or
$P \equiv Q$	logically equivalent	$\therefore$	therefore	t, c	tautology, contradiction
$\forall$	for all	$\exists$	there exists	$\exists!$	there exists exactly one
$x \in A$	x is an element of A	$A \subseteq B$	A is a subset of B	$A \subsetneq B$	proper subset
$\emptyset$	empty set	$ A $	cardinality	$\mathcal{P}(A)$	power set
$A \times B$	Cartesian product	$A \setminus B$	set difference	$\overline{A}$	complement
$A \oplus B$	symmetric difference	(a, b)	ordered pair	$\{x \in U : P(x)\}$	set-builder
$d \mid n$	d divides n	$d \nmid n$	d does not divide n	$a \equiv b \pmod{n}$	congruent mod n
xRy	$(x, y) \in R$	R^{-1}	inverse relation	$S \circ R$	R then S
R^r, R^s, R^t	closures	$[x]_{\sim}$	class of x	$A/\sim$	quotient: set of classes
$x \preceq y$	x precedes-or-equals y	$x \preceq^* y$	a linearization	$(A, \preceq)$	poset
$\mathbb{Z}^+, \mathbb{Z}_{\geq 0}$	positive / non-negative integers	$\mathbb{N}$	$\{0, 1, 2, \dots\}$	$n \operatorname{div} d, n \operatorname{mod} d$	quotient, remainder

32 · Definition index

Antichain: no two distinct elements comparable. §24

Antisymmetric: $xRy \wedge yRx \rightarrow x = y$. §21

Argument; valid; sound: premises then conclusion; valid if true premises force a true conclusion; sound if valid with true premises. §4

Asymmetric: $xRy \rightarrow y \not R x$. §21

Biconditional: $p \leftrightarrow q$: true iff p, q agree. §3

Cardinality: $|S|$, the number of elements. §13

Cartesian product: $A \times B = \{(a, b) : a \in A \wedge b \in B\}$. §15

Chain; maximal chain; length: pairwise comparable subset; cannot be extended; one less than its size. §24

Comparable: $a \preceq b$ or $b \preceq a$. §24

Compatible: $\exists c (a \preceq c \wedge b \preceq c)$. §24

Complement: $\bar{A} = U \setminus A$. §16

Composite: $n > 1$, $n = rs$ with $1 < r, s < n$. §10

Composition: $x(S \circ R)z \leftrightarrow \exists y (xRy \wedge ySz)$; R first. §20

Congruence: $a \equiv b \pmod{n} \leftrightarrow n \mid (a - b)$. §23

Contradiction: statement form false in every row. §1

Contrapositive / converse / inverse: of $p \rightarrow q$: $\sim q \rightarrow \sim p$ / $q \rightarrow p$ / $\sim p \rightarrow \sim q$. §3

Difference: $B \setminus A = \{x : x \in B \wedge x \notin A\}$. §16

div, mod: quotient and remainder of the QR theorem. §10

Divides: $d \mid n \leftrightarrow \exists k \in \mathbb{Z} (n = dk)$. §10

Domain, co-domain, range: first coordinates used; B ; second coordinates used. §19

Empty set: $\emptyset$, the set with no elements; $\emptyset \subseteq$ every set. §14

Equivalence class: $[a] = \{x \in A : a \sim x\}$. §23

Equivalence relation: reflexive, symmetric and transitive. §23

Even / odd: $n = 2k$ / $n = 2k + 1$ for some $k \in \mathbb{Z}$. §10

Existential statement: true iff some element of the domain satisfies it. §5

Hasse diagram: line x below y iff $x \preceq y$ with nothing strictly between. §24

Induced relation: same component of a partition. §23

Intersection: $A \cap B = \{x : x \in A \wedge x \in B\}$. §16

Inverse relation: $R^{-1} = \{(y, x) : (x, y) \in R\}$. §19

Largest / smallest: $\forall x (x \preceq c)$ / $\forall x (c \preceq x)$. §24

Linearization: total order $\preceq^*$ with $x \preceq y \rightarrow x \preceq^* y$. §25

Logical equivalence: identical truth values in every row. §1

Maximal / minimal: $\forall x (c \preceq x \rightarrow c = x)$ / $\forall x (x \preceq c \rightarrow c = x)$. §24

Necessary / sufficient: r necessary for s : $s \rightarrow r$; sufficient: $r \rightarrow s$. §3

Only if: p only if q : $p \rightarrow q$. §3

Ordered pair: $(a, b) = (c, d) \leftrightarrow a = c \wedge b = d$. §15

Partial order; poset: reflexive, antisymmetric, transitive; $(A, \preceq)$. §24

Partition: non-empty subsets; each element in exactly one. §18

Power set: $\mathcal{P}(A)$, the set of all subsets of A . §18

Predicate; truth set: sentence with variables; $\{x \in D : P(x)\}$. §5

Prime: $n > 1$ whose only positive factorizations are $1 \cdot n, n \cdot 1$. §10

Proper subset: $A \subseteq B$ and $A \neq B$. §14

Quotient $A/\sim$: $\{[x] : x \in A\}$. §23

Rational: $r = a/b$ with $a, b \in \mathbb{Z}, b \neq 0$. §10

Reflexive: $\forall x (xRx)$. §21

Relation: subset of $A \times B$; on A : subset of $A \times A$. §19

Replacement notation: $\{t(x) : x \in A\}$. §13

Set-builder notation: $\{x \in U : P(x)\}$. §13

Statement; statement form: true or false, not both; built from variables and connectives. §1

Subset: $\forall x (x \in A \rightarrow x \in B)$. §14

Symmetric: $xRy \rightarrow yRx$. §21

Symmetric difference: $A \oplus B = (A \setminus B) \cup (B \setminus A)$. §16

Tautology: statement form true in every row. §1

Total order: partial order in which every two elements are comparable. §24

Transitive: $xRy \wedge yRz \rightarrow xRz$. §21

Transitive closure: smallest transitive relation containing R . §22

Union: $A \cup B = \{x : x \in A \vee x \in B\}$. §16

Universal statement: true iff every element of the domain satisfies it. §5

Vacuous truth: $\forall x (P(x) \rightarrow Q(x))$ with no x satisfying P . §6

Well-ordered: every non-empty subset has a smallest element. §24